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Modeling tissue co-regulation estimates tissue-specific contributions to disease

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Integrative analyses of genome-wide association studies and gene expression data have implicated many disease-critical tissues. However, co-regulation of genetic effects on gene expression across tissues impedes distinguishing biologically causal tissues from tagging tissues. In the present study, we introduce tissue co-regulation score regression (TCSC), which disentangles causal tissues from tagging tissues by regressing gene–disease association statistics (from transcriptome-wide association studies) on tissue co-regulation scores, reflecting correlations of predicted gene expression across genes and tissues. We applied TCSC to 78 diseases/traits (average n = 302,000) and gene expression prediction models for 48 GTEx tissues. TCSC identified 21 causal tissue–trait pairs at a 5% false discovery rate (FDR), including well-established findings, biologically plausible new findings (for example, aorta artery and glaucoma) and increased specificity of known tissue–trait associations (for example, subcutaneous adipose, but not visceral adipose, and high-density lipoprotein). TCSC also identified 17 causal tissue–trait covariance pairs at 5% FDR. In conclusion, TCSC is a precise method for distinguishing causal tissues from tagging tissues.

Most diseases are driven by tissue-specific or cell-type-specific mechanisms, thus the inference of causal disease tissues is an important goal[1]. For many polygenic diseases and complex traits, disease-associated tissues have previously been identified via the integration of genome-wide association studies (GWASs) with tissue-level functional data characterizing expression quantitative trait loci (eQTLs)[2–5], gene expression[6–9] or epigenetic features[10–17]. However, it is likely that most disease-associated tissues are not actually causal, due to the high correlation of eQTL effects (respectively, gene expression or epigenetic features) across tissues; the correlation of eQTL effects across tissues, that is, tissue co-regulation, can arise due to shared eQTLs or distinct eQTLs in linkage disequilibrium (LD)[2,5,18,19]. One approach to address this involves comparing eQTL–disease colocalizations across different tissues[2] whereas another approach leverages multi-trait fine-mapping to simultaneously evaluate all tissues for disease colocalization[5]. Both methods risk implicating co-regulated tagging tissues that colocalize with disease. To our knowledge, no previous study has formally modeled genetic co-regulation across tissues to statistically disentangle causal from tagging tissues.

In the present study, we introduce a new method, TCSC, which disentangles causal tissues from tagging tissues and partitions disease heritability (or genetic covariance of two diseases/traits) into tissue-specific components. TCSC leverages gene–disease association statistics across tissues from transcriptome-wide association studies (TWASs)[18,20,21]. A challenge is that TWAS association statistics include the effects of both co-regulated tissues (see above) and co-regulated genes[18,22]. To address this, TCSC regresses TWAS χ^2 statistics (or products of z-scores for two diseases/traits) on tissue co-regulation scores, reflecting correlations of predicted gene expression across genes and tissues. Distinct from previous methods that analyze each tissue marginally, TCSC jointly models contributions from each tissue to identify causal tissues (analogous to the distinction in GWASs between marginal association and fine-mapping[23]). We validated TCSC using extensive simulations and real genotypes with LD, including comparisons to RTC

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Coloc[2], RolyPoly[6], LDSC-SEG[7] and CoCoNet[9] (reviewed in refs. 1,24). We applied TCSC to 78 diseases and complex traits (average n = 302,000) and 48 Genotype-Tissue Expression (GTEx) tissues[19], showing that TCSC recapitulates known biology and identifies biologically plausible new tissue–trait pairs (or tissue–trait covariance pairs) while attaining increased specificity relative to previous methods.

Results

Overview of TCSC regression

TCSC estimates the disease heritability explained by *cis*-genetic components of gene expression in each tissue when jointly modeling contributions from each tissue. We refer to tissues with nonzero contributions as 'causal' tissues (with the caveat that joint-fit effects of gene expression on disease may not reflect biological causality; see Discussion). TCSC assumes that gene expression–disease effect sizes are independent and identically distributed (i.i.d.) across genes and tissues (while accounting for the fact that *cis*-genetic components of gene expression are correlated across genes and tissues); violations of this model assumption are explored via simulations below. TCSC leverages the fact that TWAS χ^2 statistics for each gene and tissue include both causal effects of that gene and tissue on disease and tagging effects of co-regulated genes and tissues. We define co-regulation based on squared correlations in *cis*-genetic expression, which can arise due to shared causal eQTLs and/or LD between causal eQTLs[18]. TCSC determines that a tissue is causal for disease if genes and tissues with high co-regulation to that tissue have higher TWAS χ^2 statistics than genes and tissues with low co-regulation to that tissue.

In detail, let $h^2_{\mathrm{ge}(t')}$ denote the disease heritability explained by the *cis*-genetic component of gene expression in tissue t'. The expected TWAS χ^2 statistic for gene g and tagging tissue t is:

$$E\left[\chi^2_{g,t}\right] = N\sum_{t'} l(g,t;t')\, h^2_{\mathrm{ge}(t')}/G_{t'} + 1, \qquad (1)$$

where N is GWAS sample size, t' indexes causal tissues, $l(g,t;t')$ is a tissue co-regulation score (defined as $l(g,t;t') = \sum_{g'} r^2\left(\hat{W}_{g,t}, W_{g',t'}\right)$, where W denotes the *cis*-genetic component of gene expression for a gene–tissue pair across individuals, $\hat{W}$ denotes the *cis*-predicted expression for a gene–tissue pair, the sum is over genes g' within ±1 Mb of gene g), and $G_{t'}$ is the number of significantly *cis*-heritable genes in tissue t'. A derivation of equation (1) is provided in Methods. Equation (1) allows us to estimate $h^2_{\mathrm{ge}(t')}$ via a multiple linear regression of TWAS χ^2 statistics (for each gene and tagging tissue) on tissue co-regulation scores (Fig. 1).

TCSC can also estimate the genetic covariance between two diseases explained by *cis*-genetic components of gene expression in each tissue, using products of TWAS z-scores. In detail, let $\omega_{\mathrm{ge}(t')}$ denote the genetic covariance explained by the *cis*-genetic component of gene expression in tissue t' (Methods). The expected product of TWAS z-scores in disease 1 and disease 2 for gene g and tagging tissue t is:

$$E\left[z^1_{gt} \times z^2_{gt}\right] = \sqrt{N_1 N_2}\sum_{t'} l(g,t;t')\,\omega_{\mathrm{ge}(t')}/G_{t'} + \rho N_s/\sqrt{N_1 N_2} \qquad (2)$$

where N_1 is GWAS sample size for disease 1, N_2 is GWAS sample size for disease 2, t' indexes causal tissues, $l(g,t;t')$ is a tissue co-regulation score, $G_{t'}$ is the number of significantly *cis*-heritable genes in tissue t' (Methods), ρ is the phenotypic correlation between disease 1 and disease 2, and N_s is the number of overlapping GWAS samples between disease 1 and disease 2. Equation (2) allows us to estimate $\omega_{\mathrm{ge}(t')}$ via a multiple linear regression of products of TWAS z-scores in disease 1 and disease 2 (for each gene and tagging tissue) on tissue co-regulation scores. We note that the last term in equation (2) is not known a priori but is accounted for via the regression intercept, analogous to previous work[25].

Further details, including a formal definition of $h^2_{\mathrm{ge}(t')}$ in terms of SNP-level effects, quality control of gene expression models and TWAS statistics, estimation of jackknife standard errors, correcting for bias in tissue co-regulation scores (analogous to gene co-regulation score regression (GCSC)[22]), are provided in Methods. We have publicly released open-source software implementing TCSC regression (Code availability), as well as all GWAS summary statistics, TWAS association statistics, tissue co-regulation scores and TCSC output from the present study (Data availability).

Simulations

We performed extensive simulations to evaluate the robustness and power of TCSC, using the TWAS simulator of Mancuso et al.[26] (Code availability). We used real genotypes from the 1000 Genomes project[27] to simulate gene expression values (for each gene and tissue) and complex trait phenotypes, and then computed TWAS association statistics for each gene and tissue and estimated co-regulation scores. Primary simulations partitioned trait heritability across ten highly correlated tissues (one of which contained causal gene–trait effects explaining 100% of trait heritability). We evaluated TCSC while varying many parameters including trait heritability, explained by the causal tissue, trait heritability, not explained by gene expression, and the number of causal tissues and the number of tagging tissues; more details are provided in section 'Simulation framework' in the Supplementary Note. We also compared the type I error and power of TCSC with four previously published methods: RTC Coloc[2], RolyPoly[6], LDSC-SEG[7] and CoCoNet[9].

We first evaluated the bias in TCSC estimates of the disease heritability explained by the *cis*-genetic component of gene expression in tissue t' ($h^2_{\mathrm{ge}(t')}$), for both causal and noncausal tissues. For causal tissues, TCSC produced unbiased estimates of $h^2_{\mathrm{ge}(t')}$ (Fig. 2a and Supplementary Table 1); this implies that error in eQTL effect size estimates, which impacts TWAS statistics and co-regulation scores, does not bias TCSC estimates for causal tissues. A subtlety is that, as noted above, estimates of $h^2_{\mathrm{ge}(t')}$ are impacted by the number of significantly *cis*-heritable genes in tissue t' ($G_{t'}$), which may be sensitive to eQTL sample size. For noncausal tissues, TCSC produced estimates of $h^2_{\mathrm{ge}(t')}$ that were significantly positive when averaged across all simulations, but not large enough to substantially impact type I error (see below).

We next evaluated the type I error of TCSC for noncausal tissues. The type I error of TCSC was approximately well calibrated, ranging from 5.2% to 6.9% across eQTL sample sizes of 100–1,500 at a significance threshold of P = 0.05 (Fig. 2b and Supplementary Table 1). In comparison, we observed type I errors from 53% to 86% for RTC Coloc, 32% to 33% for LDSC-SEG, 11% to 12% for RolyPoly and 32% to 38% for CoCoNet, substantially greater than the type I error of TCSC (Extended Data Fig. 1 and Supplementary Table 2). We confirmed that the type I error of TCSC does not increase further at larger gene expression sample sizes (Fig. 2c and the section 'Single-trait simulation analysis at large sample size' in the Supplementary Note).

We next evaluated the power of TCSC for causal tissues. We determined that TCSC was moderately well powered to detect causal tissues, with power ranging from 11% to 49% across eQTL sample sizes at a nominal significance threshold of $P < 0.05$ (Fig. 2d and Supplementary Table 1). As noted above, the power of TCSC varies greatly with the choice of parameter settings (see below), so the power of TCSC in real-world settings is best evaluated using real trait analysis. As expected, power increased at larger eQTL sample sizes, due to lower s.e. values on point estimates of $h^2_{\mathrm{ge}(t')}$ (Fig. 2a). We also evaluated the power of RTC Coloc, RolyPoly, LDSC-SEG and CoCoNet. For the only other method with a type I error <15% (RolyPoly), power ranged from 14% to 17% across eQTL sample sizes, substantially lower than TCSC (Extended Data Fig. 1 and Supplementary Table 2). We also used receiver operating characteristic curves to assess the relationship between the sensitivity (power) and specificity (1 − false-positive rate) of all 5 methods across 1,000 uniformly spaced P-value thresholds. TCSC attained the largest area under the curve (AUC; 0.78, versus 0.54–0.59 for other

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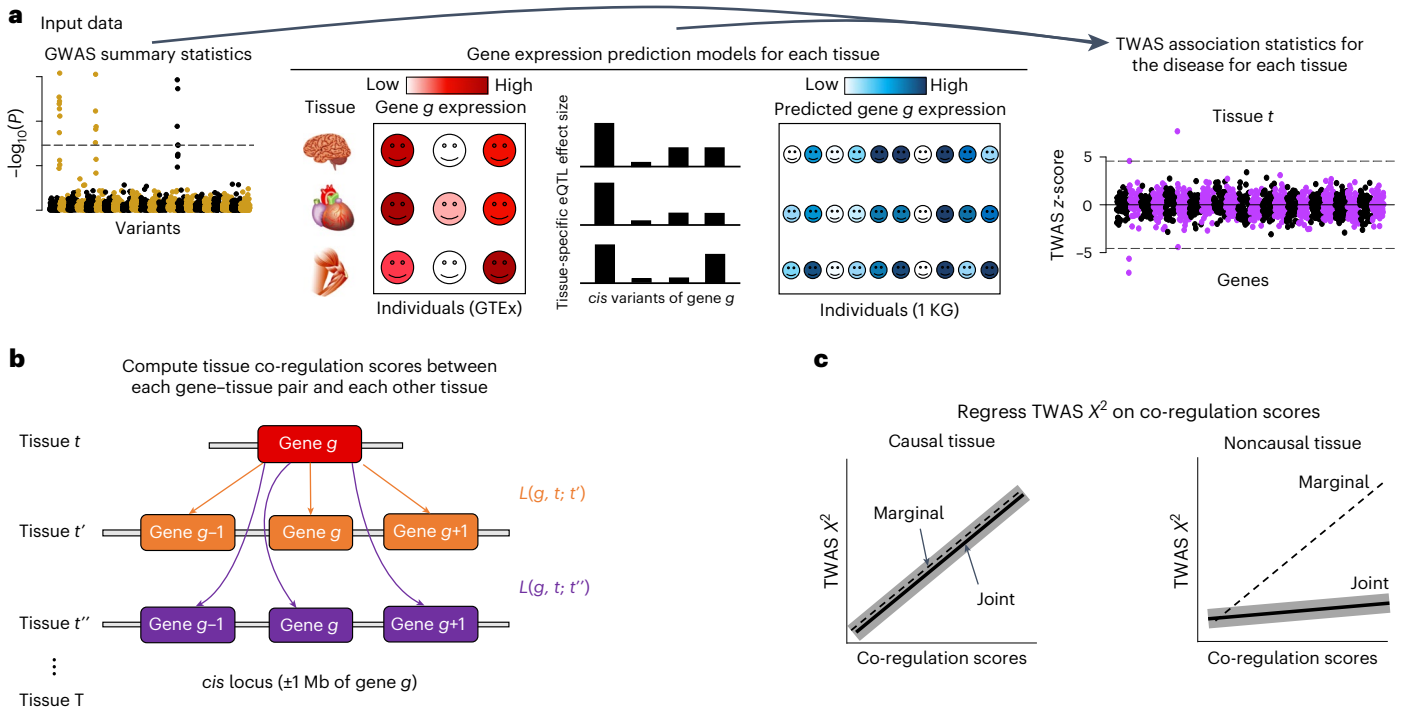


Fig. 1 | Overview of TCSC regression. a, Input data to TCSC includes (1) GWAS summary statistics for a disease and (2) gene expression prediction models for each tissue, which are used to produce (3) TWAS summary statistics for the disease for each tissue; 1000 Genomes (1KG) data used as a reference population. **b**, TCSC computes tissue co-regulation scores $L(g, t; t')$ for each gene–tissue pair (g, t) with potentially causal tissues t'. **c**, TCSC regresses TWAS χ^2 on tissue co-regulation scores to estimate tissue-specific contributions to disease. Solid lines represent hypothetical TCSC estimates and dashed lines hypothetical TWAS associations; shadows represent hypothetical TCSC estimates ± jackknife s.e. values (joint models only).

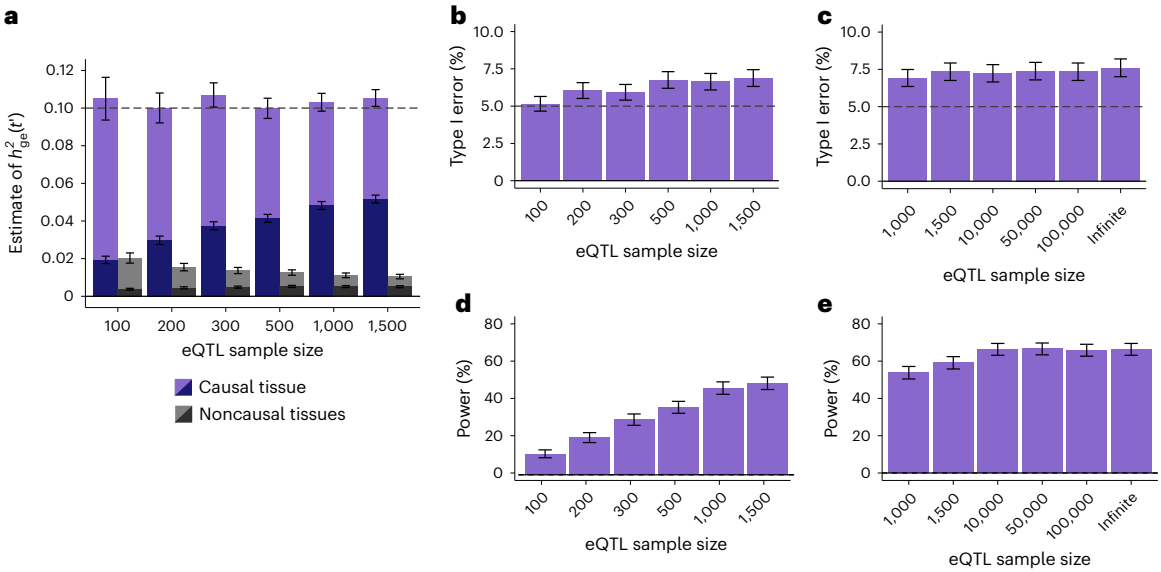


Fig. 2 | Robustness and power of TCSC regression in simulations. a, Bias in estimates of disease heritability explained by the *cis*-genetic component of gene expression in tissue t' ($h^2_{ge(t')}$) for causal (purple) and noncausal (gray) tissues. $G_{t'}$ was set to the total number of true *cis*-heritable genes across tissues for light purple and light gray, or to the number of significantly *cis*-heritable genes in each tissue for dark purple and dark gray. The dashed line indicates the true value of $h^2_{ge(t')}$ for causal tissues. **b**, Percentage of estimates of $h^2_{ge(t')}$ for noncausal tissues that were significantly positive at $P < 0.05$ per eQTL sample size ($n = 100, 200, 300, 500, 1{,}000, 1{,}500$). **c**, Percentage of estimates of $h^2_{ge(t')}$ for noncausal tissues that were significantly positive at $P < 0.05$ per larger eQTL sample size ($n = 1{,}000, 1{,}500, 10{,}000, 50{,}000, 100{,}000$, infinite). **d**, Percentage of estimates of $h^2_{ge(t')}$ for causal tissues that were significantly positive at $P < 0.05$ per eQTL sample size ($n = 100, 200, 300, 500, 1{,}000, 1{,}500$). **e**, Percentage of estimates of $h^2_{ge(t')}$ for causal tissues that were significantly positive at $P < 0.05$ per larger eQTL sample size ($n = 1{,}000, 1{,}500, 10{,}000, 50{,}000, 100{,}000$, infinite). For **b**–**e**, we used a one-sided z-test to obtain P values. For **a**, **b** and **d**, we performed $n = 1{,}000$ independent simulations per eQTL sample size. For **c** and **e**, we performed $n = 2{,}000$ independent simulations per eQTL sample size and used cross-validation adjusted $R^2 > 0$ instead of GCTA to define significantly *cis*-heritable genes in these analyses. In all panels, data are presented as mean values ± 1.96 × s.e. The numerical results are reported in Supplementary Table 1.

methods) (Extended Data Fig. 1). We also evaluated the power of TCSC at 4 larger gene expression sample sizes (see above) and determined that power plateaus at 10,000 individuals (Fig. 2e).

We similarly evaluated cross-trait TCSC for simulated traits with a genetic correlation of 0.5. Cross-trait simulations produced similar conclusions to the single-trait analysis above (Extended Data Figs. 1 and 2, Supplementary Table 3 and Supplementary Note). We also performed 12 secondary simulation analyses showing the performance of TCSC across a diverse parameter space (Extended Data Figs. 3–6, Supplementary Figs. 2–21 and Supplementary Note).

Tissue-specific contributions to 78 diseases/traits

We applied TCSC to publicly available GWAS summary statistics for 78 diseases and complex traits (average n = 302,000; Methods and Supplementary Table 4) and gene expression data for 48 GTEx tissues[19] (Table 1). The 48 GTEx tissues were aggregated into 39 meta-tissues (average n = 266, range: n = 101–320 individuals, 23 meta-tissues with n = 320) to reduce variation in eQTL sample size across tissues (Table 1 and the section 'Analyzing GTEx tissues' in the Supplementary Note) and referred to as 'tissues' below. We constructed gene expression prediction models for an average of 3,993 significantly *cis*-heritable, protein-coding genes (as defined above) in each tissue. We primarily report the proportion of disease heritability explained by the *cis*-genetic component of gene expression in tissue t' ($\pi_{t'} = h^2_{ge(t')}/h^2_g$), where h^2_g is the common variant SNP heritability estimated by S-LDSC[13,28,29].

TCSC identified 21 causal tissue–trait pairs with significantly positive contributions to disease/trait heritability at 5% per-trait FDR, spanning 7 distinct tissues and 17 distinct diseases/traits (Fig. 3, Methods, Supplementary Table 5 and Supplementary Fig. 22). Many of the significant findings recapitulated known biology, including associations of whole blood with blood cell traits such as white blood cell (WBC) count ($\pi_{t'}$ = 0.21, s.e. = 0.064, $P = 5.7 \times 10^{-4}$) and liver with lipid traits such as low-density lipoprotein (LDL; $\pi_{t'}$ = 0.20, s.e. = 0.050, $P = 2.9 \times 10^{-5}$). TCSC also identified several biologically plausible findings not previously reported in the genetics literature. First, the aortic artery was associated with glaucoma ($\pi_{t'}$ = 0.15, s.e. = 0.051, $P = 1.3 \times 10^{-3}$) and also with diastolic blood pressure (DBP) ($\pi_{t'}$ = 0.078, s.e. = 0.024, $P = 5.1 \times 10^{-4}$), which is consistent with DBP measuring the pressure exerted on the aorta[30] and high blood pressure being a known risk factor for glaucoma[31–35]. Second, TCSC identified the left ventricle of the heart as a causal tissue for platelet count ($\pi_{t'}$ = 0.091, s.e. = 0.031, $P = 1.7 \times 10^{-3}$), consistent with the role of platelets in the formation of blood clots in cardiovascular disease[36–39]; in fact, antiplatelet drugs have been successful at reducing adverse cardiovascular outcomes[40]. Moreover, the left ventricle serves as a muscle to pump blood throughout the body[41], probably modulating platelet and other blood cell counts. Other significant findings are reported in Supplementary Table 6 and Supplementary Note.

TCSC also increased the specificity of known tissue–trait associations. For high-density lipoprotein (HDL), previous studies reported that deletion of a cholesterol transporter gene in adipose tissue reduces HDL levels[42,43]. TCSC specifically identified subcutaneous adipose ($\pi_{t'}$ = 0.16, s.e. = 0.054, $P = 1.5 \times 10^{-3}$; Fig. 3), but not visceral adipose or breast adipose tissue ($P > 0.05$; Supplementary Table 6), as a causal tissue for HDL. Previous studies have established that levels of adiponectin, a hormone released by adipose tissue to regulate insulin, are positively correlated with HDL[44–46] and, more recently, a study has reported that adiponectin levels are associated specifically with subcutaneous adipose tissue and not visceral adipose tissue[47]. We note that TCSC did not identify liver as a causal tissue for HDL (FDR > 5%), which may be due to limited power in liver as a result of a smaller eQTL sample size. For waist:hip ratio adjusted for body mass index (BMI) (WHRadjBMI), previous studies reported colocalization of WHRadjBMI GWAS variants with *cis*-eQTLs in subcutaneous adipose, visceral adipose, liver and whole blood[48], consistent with WHRadjBMI measuring adiposity in the intra-abdominal space, which is probably regulated by metabolically active tissues[49]. TCSC specifically identified subcutaneous adipose as a suggestive finding ($\pi_{t'}$ = 0.10, s.e. = 0.037, $P = 2.4 \times 10^{-3}$,

Table 1 | GTEx meta-tissues and constituent tissues analyzed

Meta-tissue	Constituent tissue(s)	Sample size
Adipose subcutaneous	Adipose subcutaneous	320
Adipose visceral omentum	Adipose visceral omentum	320
Adrenal gland	Adrenal gland	200
Aortic artery	Aortic artery	320
†Brain basal ganglia	Brain putamen, caudate, nucleus accumbens	320
†Brain cerebellum	Brain cerebellum, cerebellar hemisphere	320
†Brain cortex	Brain frontal, anterior, cingulate	320
†Brain limbic	Brain amygdala, hippocampus, hypothalamus	320
Brain spinal cord	Brain spinal cord	115
Brain substantia nigra	Brain substantia nigra	101
Breast, mammary gland	Breast mammary gland	320
Coronary artery	Coronary artery	180
Cultured fibroblasts	Cultured fibroblasts	320
Esophagus mucosa	Esophagus mucosa	320
Esophagus muscularis	Esophagus muscularis	320
Heart atrial appendage	Heart atrial appendage	320
Heart left ventricle	Heart left ventricle	320
LCLs	LCLs	116
Liver	Liver	183
Lung	Lung	320
Minor salivary gland	Minor salivary gland	118
Muscle skeletal	Muscle skeletal	320
Ovary	Ovary	140
Pancreas	Pancreas	252
Pituitary	Pituitary	220
Prostate	Prostate	186
Skin (sun exposed)	Skin (sun exposed)	320
Skin (sun unexposed)	Skin (sun unexposed)	320
†Sigmoid intestine	Sigmoid colon, gastroesophageal junction	320
Spleen	Spleen	185
Stomach	Stomach	269
Tibial artery	Tibial artery	320
Tibial nerve	Tibial nerve	320
Testis	Testis	277
Thyroid	Thyroid	320
†Transverse intestine	Transverse colon, small intestine	320
Uterus	Uterus	108
Vagina	Vagina	122
Whole blood	Whole blood	320

For each meta-tissue, we list the constituent tissue(s) and total sample size. The dagger denotes meta-tissues with more than one constituent tissue; for these meta-tissues, each constituent tissue has equal sample size up to rounding error (an exception is the transverse intestine meta-tissue, which includes 176 transverse colon samples and all 144 small intestine samples).

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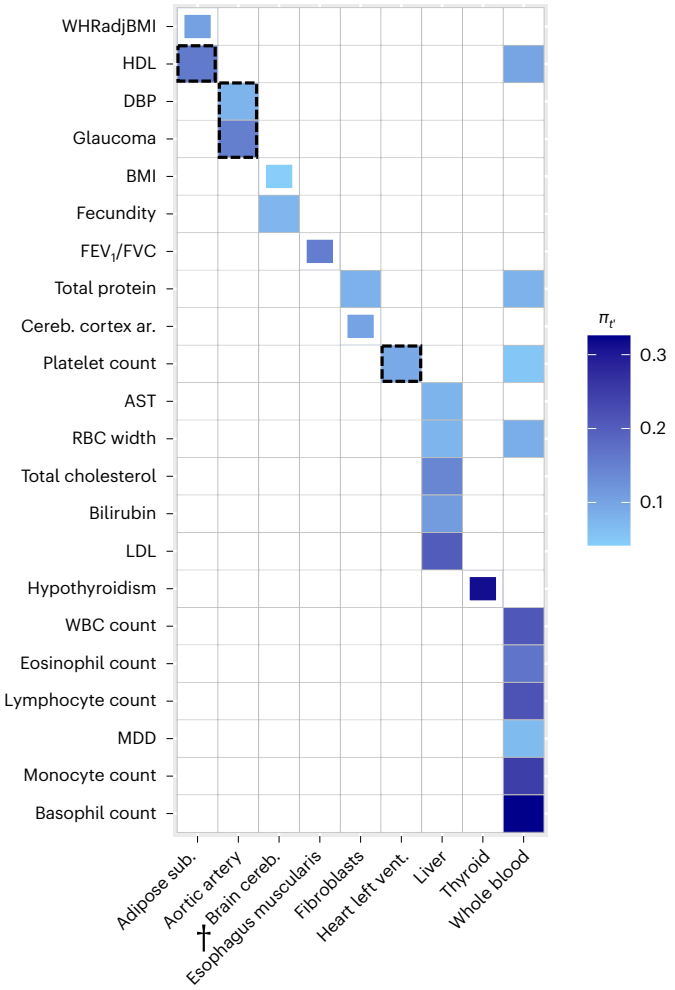


Fig. 3 | TCSC estimates tissue-specific contributions to disease and complex trait heritability. We report estimates of the proportion of disease heritability explained by the *cis*-genetic component of gene expression in tissue t' ($\pi_{t'}$). We report tissue–trait pairs with an FDR ≤10%, where full boxes denote FDR ≤5% and partial boxes FDR between 5% and 10%. The dashed boxes denote the results that are highlighted in the main text. Tissues are ordered alphabetically. The dagger denotes meta-tissues with more than one constituent tissue. Diseases/traits are ordered with respect to causal tissues. The numerical results are reported in Supplementary Tables 5 and 6. Adipose sub., Adipose, subcutaneous; AST, Aspartate aminotransferase; Brain cereb., Brain cerebellum; Cereb. cortex ar., Cerebral cortex surface area; FEV_1/FVC, forced expiratory volume in 1 second divided by forced vital capacity; MDD: major depressive disorder; vent., ventricle.

5% < FDR < 10%; Fig. 3), but not visceral adipose, breast, liver or whole blood ($P > 0.05$; Supplementary Table 6). The causal mechanism may again involve adiponectin, which is negatively correlated with WHRadjBMI[50]. We note that the *P*-value distributions across traits are similar for adipose tissues (Supplementary Table 7). For BMI, previous studies have broadly implicated the central nervous system, but did not reveal more precise contributions[7,13,51–54]. TCSC specifically identified brain cerebellum as a suggestive finding ($\pi_{t'} = 0.042$, s.e. = 0.015, $P = 2.6 \times 10^{-3}$, 5% < FDR < 10%), but not brain cortex or brain limbic ($P > 0.05$; Supplementary Table 6), as a causal tissue for BMI. This finding is consistent with a known role for brain cerebellum in biological processes related to obesity, including endocrine homeostasis[55] and feeding control[56]; recently, a multi-omics approach has revealed cerebellar activation in mice when feeding[57].

We performed several secondary analyses. First, we removed tissues with eQTL sample size <320 individuals, because these tissues may often be underpowered (Fig. 2c). We identified 23 significant tissue–trait pairs, reflecting a gain of 8 newly significant tissue–trait pairs (Extended Data Fig. 7, Supplementary Table 8 and Supplementary Note). We also performed a brain-specific analysis in which we applied TCSC to 41 brain traits (average $n = 226{,}000$) while restricting to 13 individual GTEx brain tissues (Supplementary Tables 10 and 11 and Supplementary Note). TCSC identified eight brain tissue–brain trait pairs at 5% FDR, including the brain hippocampus as a causal tissue for attention deficit hyperactivity disorder (ADHD) consistent with dopamine regulation[58–60] (Extended Data Fig. 8 and Supplementary Table 12).

Comparisons of TCSC to other methods

We compared TCSC with two methods, RTC Coloc[2] and LDSC-SEG[7], that use gene expression data to identify disease-critical tissues based on tissue specificity of eQTL-GWAS colocalizations or heritability enrichment of specifically expressed genes, respectively. Both methods analyze each tissue marginally; thus, RTC Coloc and LDSC-SEG may output more significantly associated tissues per trait than TCSC (analogous to the distinction in GWASs between marginal association and fine-mapping[23]). To assess whether TCSC attains higher specificity, we applied each method to seven traits with at least one tissue–trait association for each of the three methods (Methods), and compared results across causal tissues identified by TCSC and across the most strongly co-regulated tagging tissue (based on Spearman's ρ for estimated eQTL effect sizes, averaged across genes, from ref. 19).

Results for the 7 traits are reported in Fig. 4 and Supplementary Table 13 and those for all 17 diseases/traits with causal tissue–trait associations identified by TCSC are reported in Extended Data Fig. 9 and Supplementary Table 14; the results for all diseases/traits and tissues are in Supplementary Table 15. We reached three main conclusions: first, RTC Coloc implicates a broad set of tissues (not just strongly co-regulated tissues) for each trait (Fig. 4a), for example, for WBC count, RTC Coloc implicated eight of ten tissues. This is consistent with our simulations, in which RTC Coloc suffered a high type I error rate and had a substantially lower AUC than TCSC (Extended Data Fig. 1). Second, LDSC-SEG implicates a small set of strongly co-regulated tissues for each trait (Fig. 4b); for WBC count, LDSC-SEG typically implicated three of ten tissues, consisting of whole blood, spleen and breast tissue. This is consistent with our simulations, in which LDSC-SEG suffered a substantial type I error rate and had a substantially lower AUC than TCSC (Extended Data Fig. 1). Third, TCSC typically implicates one causal tissue per trait (Fig. 4c): for WBC count, TCSC implicated only whole blood as a causal tissue. This is consistent with our simulations, in which TCSC attained moderate power to identify causal tissues with an approximately well-calibrated type I error. However, the higher specificity of TCSC may be accompanied by incomplete power to identify secondary causal tissues; we observed less significant (lower $-\log_{10}(P)$ and lower $-\log_{10}(\text{FDR})$) results for causal tissues from TCSC than by RTC Coloc or LDSC-SEG (Supplementary Table 14). We also observed similar patterns when comparing TCSC with RTC Coloc and LDSC-SEG in the brain-specific analysis (Extended Data Fig. 10, Supplementary Table 16 and Supplementary Note).

Tissue-specific contributions to cross-trait covariance

We applied cross-trait TCSC to 262 pairs of disease/traits (for example, pairs with significantly nonzero genetic correlation; Methods and Supplementary Table 17) and gene expression data for 39 meta-tissues (Table 1). We primarily report the signed proportion of genetic covariance explained by the *cis*-genetic component of gene expression in tissue t' ($\zeta_{t'} = \omega_{ge(t')}/\omega_g$), where ω_g is the common variant genetic covariance estimated by cross-trait LDSC[61]. TCSC identified 17 causal tissue–trait covariance pairs with significant contributions to trait covariance at 5% per-trait FDR, spanning 12 distinct tissues and 13 distinct trait pairs (Fig. 5a and Supplementary Table 18).

Several results recapitulated known biology. Consistent with the significant contribution of liver to LDL heritability in Fig. 3, TCSC

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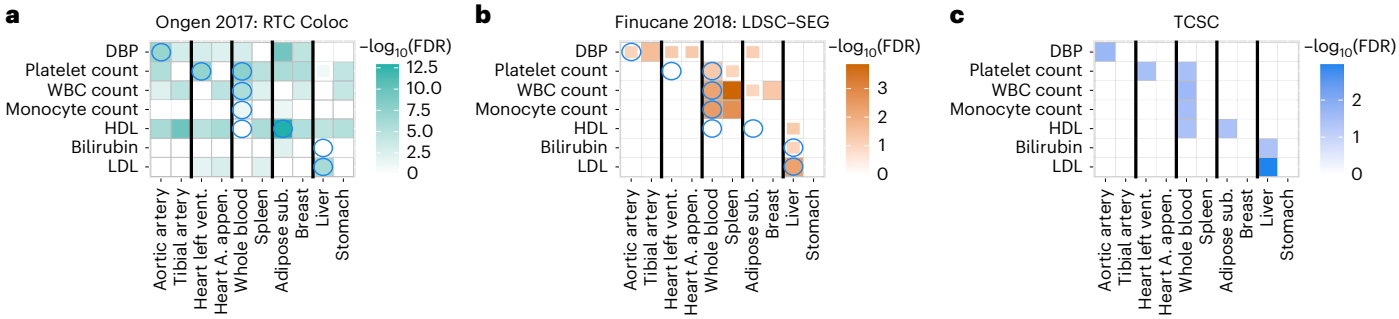


Fig. 4 | Comparison of disease-critical tissues identified by RTC Coloc, LDSC-SEG and TCSC. a–c, The $-\log_{10}$(FDR) values for RTC Coloc (**a**), LDSC-SEG (**b**) and TCSC (**c**), across seven traits with at least one significantly associated tissue (at FDR = 5%) for each of the three methods and ten tissues consisting of the causal tissues identified by TCSC and the most strongly co-regulated tagging tissues, ordered consecutively. We reported tissue–trait pairs with FDR ≤ 10%, where full boxes denote FDR ≤ 5% and partial boxes FDR between 5% and 10%. Blue circles in **a** and **b** denote the causal tissue–trait pairs identified by TCSC. Heart A. appen., Heart atrial appendage. The numerical results are reported in Supplementary Table 13.

identified a suggestive positive contribution of liver to the covariance of LDL and total cholesterol ($\zeta_{t'} = 0.090$, s.e. = 0.029, $P = 1.0 \times 10^{-3}$, 5% < FDR < 10%) and, consistent with the positive contributions of whole blood to blood cell count heritabilities in Fig. 3, TCSC identified a significant positive contribution of whole blood to the covariance of eosinophil and platelet counts ($\zeta_{t'} = 0.30$, s.e. = 0.10, $P = 2.3 \times 10^{-3}$).

TCSC identified several biologically plausible findings not previously reported in the genetics literature. First, brain substantia nigra had a significantly positive contribution to the genetic covariance of BMI and red blood cell (RBC) count ($\zeta_{t'} = 0.28$, s.e. = 0.084, $P = 4.6 \times 10^{-4}$), whereas the pancreas had a significantly negative contribution ($\zeta_{t'} = -0.25$, s.e. = 0.079, $P = 8.7 \times 10^{-4}$). In the brain, energy metabolism is regulated by oxidation and previous work has shown that RBCs play a large role in these metabolic processes as oxygen sensors[62]; in addition, previous studies have reported differences in the level of oxidative enzymes in RBCs between individuals with a high BMI and those with a low BMI[63,64], suggesting that genes regulating oxidative processes might have pleiotropic effects on RBC count and BMI. In the pancreas, pancreatic inflammation (specifically acute pancreatitis) is associated with reduced levels of RBCs, or anemia[65], whereas pancreatic fat is associated with metabolic disease and increased BMI[66]. Second, lymphoblastoid cell lines (LCLs) had a suggestive negative contribution to the genetic covariance of eosinophil count and WBC count ($\zeta_{t'} = -0.081$, s.e. = 0.028, $P = 1.8 \times 10^{-3}$, 5% < FDR < 10%), in contrast to the suggestive positive contribution of whole blood ($\zeta_{t'} = 0.32$, s.e. = 0.12, $P = 2.4 \times 10^{-3}$, 5% < FDR < 10%). This is plausible because previous studies have reported the suppression of proliferation of lymphocytes (the WBC hematopoietic lineage from which LCLs are derived) by molecules secreted from eosinophils[67–69]. Other significant findings are reported in Supplementary Table 19 and Supplementary Note.

For 16 of the 17 causal tissue–trait covariance pairs, the causal tissue was nonsignificant for both constituent traits. We sought to formally assess whether tissue-specific contributions to covariance could exceed contributions to heritability. Specifically, for each causal tissue–trait covariance pair, we estimated the differences between the tissue-specific contribution to covariance ($\zeta_{t'}$) and the tissue-specific contributions to heritability for each constituent trait ($\pi_{t'}$) (and estimated s.e. values by jackknifing differences across the genome). We identified five tissue–trait covariance pairs for which these differences were statistically significant at 5% FDR for both traits and $\pi_{t'}$ was nonsignificant for both traits (marked by the double asterisk in Fig. 5a; results reported in Supplementary Table 20 and Supplementary Note). The negative contribution of the pancreas (Fig. 5b) and the positive contribution of the brain substantia nigra (Fig. 5c) to the covariance of BMI and RBC count were larger than tissue-specific contributions to heritability, which were each nonsignificant. Even in simulations, TCSC often detected tissue-specific contributions to covariance without detecting tissue-specific contributions to heritability for both traits, especially when contributions to heritability substantially differed (Supplementary Table 21).

Discussion

We developed a new method, TCSC, that disentangles causal tissues from tagging tissues and partitions disease heritability (or cross-trait covariance) into tissue-specific components. We applied TCSC to 78 diseases and complex traits and 48 GTEx tissues, identifying 21 tissue–trait pairs (and 17 tissue–trait covariance pairs) with significant tissue-specific contributions. TCSC identified biologically plausible new tissue–trait pairs, including associations of the aortic artery with glaucoma, the left ventricle of the heart with platelet count and the brain hippocampus with ADHD. TCSC also identified biologically plausible, new, tissue–trait covariance pairs, including a positive contribution of the brain substantia nigra and a negative contribution of the pancreas to the covariance of BMI and RBC count; in particular, our findings suggest that genetic covariance may reflect distinct tissue-specific contributions.

TCSC differs from previous methods in jointly modeling contributions from each tissue to disentangle causal tissues from tagging tissues (analogous to the distinction in GWAS between marginal association and fine-mapping[23]). We briefly discuss several other methods that use eQTL or gene expression data to identify disease-associated tissues. RTC Coloc identifies disease-associated tissues based on tissue specificity of eQTL-GWAS colocalizations[2]; that study made a valuable contribution in emphasizing the importance of tissue co-regulation, but did not model tissue-specific effects, such that RTC Coloc may implicate many tissues (Fig. 4a). LDSC-SEG identifies disease-critical tissues based on heritability enrichment of specifically expressed genes[7] and distinguishes a focal tissue from the set of all tissues analyzed, but does not distinguish closely co-regulated tissues (Fig. 4b). We discuss four other methods in the section 'Other tissue–trait association methods' in the Supplementary Note.

We note several limitations to our work: first, TCSC requires tissue-specific eQTL data (thus requiring genotype/gene expression data in substantial sample size), whereas some methods (LDSC-SEG[7], RolyPoly[6] and CoCoNet[9]) require gene expression data only in limited sample size. Second, joint-fit effects of gene expression on disease may not reflect biological causality; if a causal tissue or cell type is not assayed, TCSC may identify a co-regulated tissue (for example, a tissue with cell-type composition that favors a causal cell type) as causal. We anticipate that this limitation will become less severe as potentially causal tissues, cell types and contexts are more comprehensively assayed. Third, TCSC does not achieve a strict definition or

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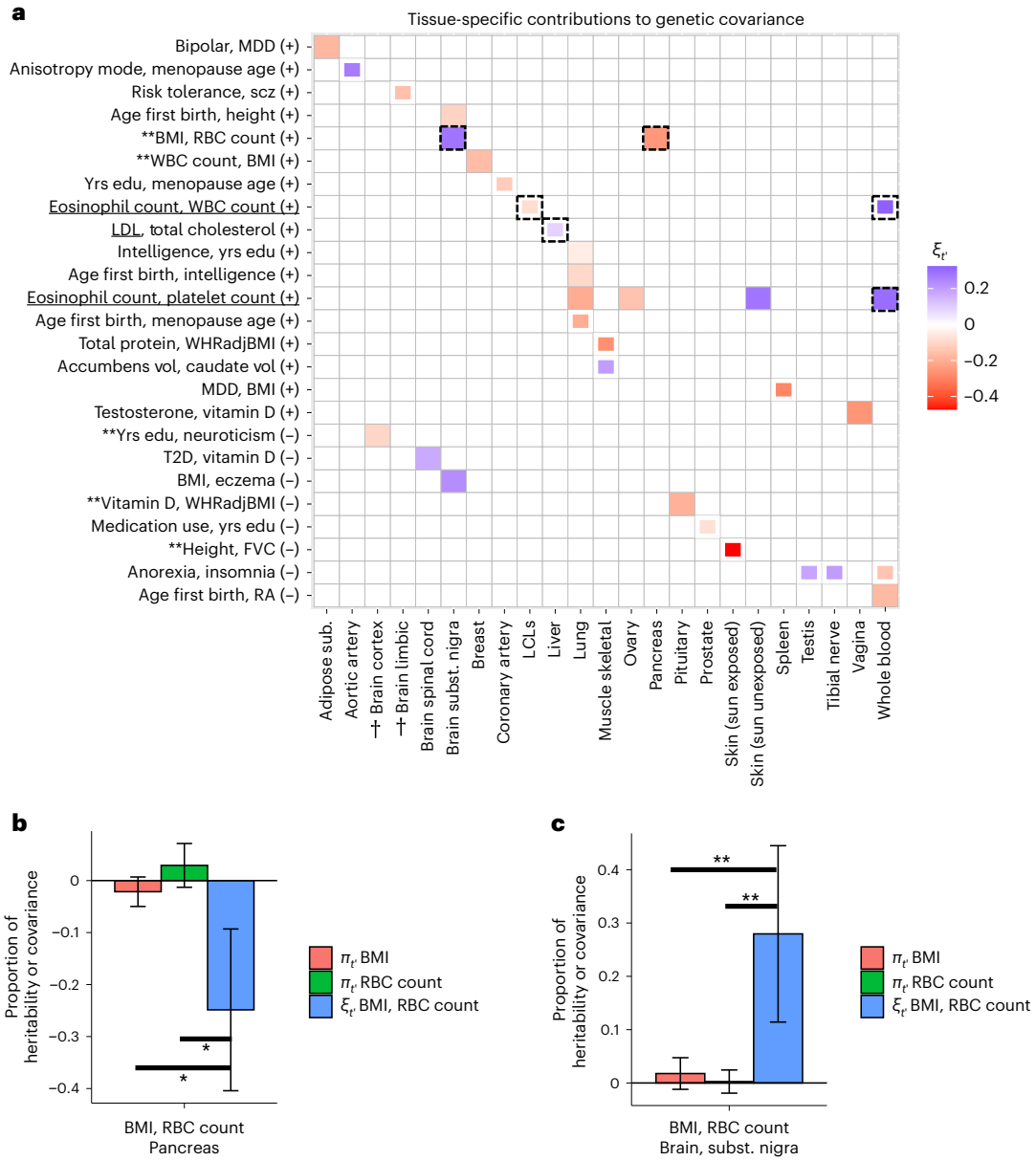


Fig. 5 | Cross-trait TCSC estimates tissue-specific contributions to the genetic covariance of two diseases/traits. a, Estimates of the signed proportion of genetic covariance explained by the *cis*-genetic component of gene expression in tissue t' ($\zeta_{t'}$). Full boxes denote FDR < 5% and partial boxes 5% < FDR < 10%. The dashed boxes are highlighted in the main text. The tissues are ordered alphabetically. The dagger denotes meta-tissues with more than one constituent tissue. Trait pairs are ordered by positive (+) or negative (−) genetic covariance and, then, with respect to causal tissues. Underlined traits are shown in Fig. 3. The double asterisk denotes trait pairs with significant differences between tissue-specific contributions to covariance and heritability, the latter of which was nonsignificant for both traits. The numerical results are reported in Supplementary Tables 18 and 19. Accumbens vol, brain accumbens volume; Caudate vol, brain caudate volume; RA, rheumatoid arthritis; Scz, schizophrenia; subst., substantia; T2D, type 2 diabetes; yrs edu, years of education. **b**, For BMI and RBC count, estimates of the proportion of trait heritability and covariance explained by the *cis*-genetic component of gene expression in the pancreas. The single asterisk denotes the difference between the covariance and the heritability estimates that is significantly <0 at 10% FDR using a genomic block jackknife to estimate the s.e. **c,** Same as **b**, but for the brain substantia nigra. The double asterisk denotes the difference between the covariance and heritability estimates significantly >0 at 5% FDR using a genomic block jackknife to estimate the s.e. Data in **b** and **c** are presented as TCSC estimates ±1.96 × the jackknife s.e. The numerical results are reported in Supplementary Table 21.

estimation of mediated effects; this is conceptually appealing and can, in principle, be achieved by modeling nonmediated effects, but may result in limited power to distinguish disease-critical tissues[70]. Fourth, TCSC performs less well in the presence of disease heritability that is not mediated through gene expression (Supplementary Figs. 16–19). We discuss other limitations in the section 'Other limitations' in the Supplementary Note. Despite these limitations, TCSC is a powerful and generalizable approach for modeling tissue co-regulation to estimate tissue-specific contributions to disease.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions

and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41588-023-01474-z.

References

1. Hekselman, I. & Yeger-Lotem, E. Mechanisms of tissue and cell-type specificity in heritable traits and diseases. *Nat. Rev. Genet.* **21**, 137–150 (2020).
2. Ongen, H. et al. Estimating the causal tissues for complex traits and diseases. *Nat. Genet.* **49**, 1676–1683 (2017).
3. Gamazon, E. R. et al. Using an atlas of gene regulation across 44 human tissues to inform complex disease- and trait-associated variation. *Nat. Genet.* **50**, 956–967 (2018).
4. Hormozdiari, F. et al. Leveraging molecular quantitative trait loci to understand the genetic architecture of diseases and complex traits. *Nat. Genet.* **50**, 1041–1047 (2018).
5. Arvanitis, M., Tayeb, K., Strober, B. J. & Battle, A. Redefining tissue specificity of genetic regulation of gene expression in the presence of allelic heterogeneity. *Am. J. Hum. Genet.* **109**, 223–239 (2022).
6. Calderon, D. et al. Inferring relevant cell types for complex traits by using single-cell gene expression. *Am. J. Hum. Genet.* **101**, 686–699 (2017).
7. Finucane, H. K. et al. Heritability enrichment of specifically expressed genes identifies disease-relevant tissues and cell types. *Nat. Genet.* **50**, 621–629 (2018).
8. Bryois, J. et al. Genetic identification of cell types underlying brain complex traits yields insights into the etiology of Parkinson's disease. *Nat. Genet.* **52**, 482–493 (2020).
9. Shang, L., Smith, J. A. & Zhou, X. Leveraging gene co-expression patterns to infer trait-relevant tissues in genome-wide association studies. *PLoS Genet.* **16**, e1008734 (2020).
10. Maurano, M. T. et al. Systematic localization of common disease-associated variation in regulatory DNA. *Science* **337**, 1190–1195 (2012).
11. Trynka, G. et al. Chromatin marks identify critical cell types for fine mapping complex trait variants. *Nat. Genet.* **45**, 124–130 (2013).
12. Pickrell, J. K. Joint analysis of functional genomic data and genome-wide association studies of 18 human traits. *Am. J. Hum. Genet.* **94**, 559–573 (2014).
13. Finucane, H. K. et al. Partitioning heritability by functional annotation using genome-wide association summary statistics. *Nat. Genet.* **47**, 1228–1235 (2015).
14. Roadmap Epigenomics Consortium et al. Integrative analysis of 111 reference human epigenomes. *Nature* **518**, 317–330 (2015).
15. Backenroth, D. et al. FUN-LDA: a latent dirichlet allocation model for predicting tissue-specific functional effects of noncoding variation: methods and applications. *Am. J. Hum. Genet.* **102**, 920–942 (2018).
16. Amariuta, T. et al. IMPACT: genomic annotation of cell-state-specific regulatory elements inferred from the epigenome of bound transcription factors. *Am. J. Hum. Genet.* **104**, 879–895 (2019).
17. Boix, C. A., James, B. T., Park, Y. P., Meuleman, W. & Kellis, M. Regulatory genomic circuitry of human disease loci by integrative epigenomics. *Nature* **590**, 300–307 (2021).
18. Wainberg, M. et al. Opportunities and challenges for transcriptome-wide association studies. *Nat. Genet.* **51**, 592–599 (2019).
19. The GTEx Consortium. The GTEx Consortium atlas of genetic regulatory effects across human tissues. *Science* **369**, 1318–1330 (2020).
20. Gamazon, E. R. et al. A gene-based association method for mapping traits using reference transcriptome data. *Nat. Genet.* **47**, 1091–1098 (2015).
21. Gusev, A. et al. Integrative approaches for large-scale transcriptome-wide association studies. *Nat. Genet.* **48**, 245–252 (2016).
22. Siewert-Rocks, K. M., Kim, S. S., Yao, D. W., Shi, H. & Price, A. L. Leveraging gene co-regulation to identify gene sets enriched for disease heritability. *Am. J. Hum. Genet.* **109**, 393–404 (2022).
23. Schaid, D. J., Chen, W. & Larson, N. B. From genome-wide associations to candidate causal variants by statistical fine-mapping. *Nat. Rev. Genet.* **19**, 491–504 (2018).
24. Zhu, H., Shang, L. & Zhou, X. A review of statistical methods for identifying trait-relevant tissues and cell types. *Front. Genet.* **11**, 587887 (2020).
25. Bulik-Sullivan, B. K. et al. LD score regression distinguishes confounding from polygenicity in genome-wide association studies. *Nat. Genet.* **47**, 291–295 (2015).
26. Mancuso, N. et al. Probabilistic fine-mapping of transcriptome-wide association studies. *Nat. Genet.* **51**, 675–682 (2019).
27. The 1000 Genomes Project Consortium. An integrated map of genetic variation from 1,092 human genomes. *Nature* **491**, 56–65 (2012).
28. Gazal, S. et al. Linkage disequilibrium-dependent architecture of human complex traits shows action of negative selection. *Nat. Genet.* **49**, 1421–1427 (2017).
29. Gazal, S., Marquez-Luna, C., Finucane, H. K. & Price, A. L. Reconciling S-LDSC and LDAK functional enrichment estimates. *Nat. Genet.* **51**, 1202–1204 (2019).
30. Homan, T. D., Bordes, S. & Cichowski, E. Physiology, pulse pressure. in *StatPearls* (Treasure Island, 2022).
31. Kass, M. A. et al. The ocular hypertension treatment study: a randomized trial determines that topical ocular hypotensive medication delays or prevents the onset of primary open-angle glaucoma. *Arch. Ophthalmol.* **120**, 701–713 (2002).
32. Zhao, D., Cho, J., Kim, M. H. & Guallar, E. The association of blood pressure and primary open-angle glaucoma: a meta-analysis. *Am. J. Ophthalmol.* **158**, 615–27 e9 (2014).
33. Levine, R. M., Yang, A., Brahma, V. & Martone, J. F. Management of blood pressure in patients with glaucoma. *Curr. Cardiol. Rep.* **19**, 109 (2017).
34. De Moraes, C. G., Cioffi, G. A., Weinreb, R. N. & Liebmann, J. M. New recommendations for the treatment of systemic hypertension and their potential implications for glaucoma management. *J. Glaucoma* **27**, 567–571 (2018).
35. Leeman, M. & Kestelyn, P. Glaucoma and blood pressure. *Hypertension* **73**, 944–950 (2019).
36. Gregg, D. & Goldschmidt-Clermont, P. J. Cardiology patient page. Platelets and cardiovascular disease. *Circulation* **108**, e88–e90 (2003).
37. Coppinger, J. A. et al. Characterization of the proteins released from activated platelets leads to localization of novel platelet proteins in human atherosclerotic lesions. *Blood* **103**, 2096–2104 (2004).
38. Gawaz, M., Langer, H. & May, A. E. Platelets in inflammation and atherogenesis. *J. Clin. Invest.* **115**, 3378–3384 (2005).
39. Davi, G. & Patrono, C. Platelet activation and atherothrombosis. *N. Engl. J. Med.* **357**, 2482–2494 (2007).
40. Meadows, T. A. & Bhatt, D. L. Clinical aspects of platelet inhibitors and thrombus formation. *Circ. Res.* **100**, 1261–1275 (2007).
41. Berman, M. N., Tupper, C. & Bhardwaj, A. Physiology, left ventricular function. in *StatPearls* (Treasure Island, 2022).
42. Chung, S., Sawyer, J. K., Gebre, A. K., Maeda, N. & Parks, J. S. Adipose tissue ATP binding cassette transporter A1 contributes to high-density lipoprotein biogenesis in vivo. *Circulation* **124**, 1663–1672 (2011).

43. McGillicuddy, F. C., Reilly, M. P. & Rader, D. J. Adipose modulation of high-density lipoprotein cholesterol: implications for obesity, high-density lipoprotein metabolism, and cardiovascular disease. *Circulation* **124**, 1602–1605 (2011).
44. Zoccali, C. et al. Adiponectin, metabolic risk factors, and cardiovascular events among patients with end-stage renal disease. *J. Am. Soc. Nephrol.* **13**, 134–141 (2002).
45. Ryo, M. et al. Adiponectin as a biomarker of the metabolic syndrome. *Circ. J.* **68**, 975–981 (2004).
46. Toth, P. P. Adiponectin and high-density lipoprotein: a metabolic association through thick and thin. *Eur. Heart J.* **26**, 1579–1581 (2005).
47. Van Linthout, S. et al. Impact of HDL on adipose tissue metabolism and adiponectin expression. *Atherosclerosis* **210**, 438–444 (2010).
48. Shungin, D. et al. New genetic loci link adipose and insulin biology to body fat distribution. *Nature* **518**, 187–196 (2015).
49. Emdin, C. A. et al. Genetic association of waist-to-hip ratio with cardiometabolic traits, type 2 diabetes, and coronary heart disease. *J. Am. Med. Assoc.* **317**, 626–634 (2017).
50. Smith, J., Al-Amri, M., Sniderman, A. & Cianflone, K. Leptin and adiponectin in relation to body fat percentage, waist to hip ratio and the apoB/apoA1 ratio in Asian Indian and Caucasian men and women. *Nutr. Metab.* **3**, 18 (2006).
51. Farooqi, I. S. Defining the neural basis of appetite and obesity: from genes to behaviour. *Clin. Med.* **14**, 286–289 (2014).
52. Locke, A. E. et al. Genetic studies of body mass index yield new insights for obesity biology. *Nature* **518**, 197–206 (2015).
53. Medic, N. et al. Increased body mass index is associated with specific regional alterations in brain structure. *Int. J. Obes.* **40**, 1177–1182 (2016).
54. Yengo, L. et al. Meta-analysis of genome-wide association studies for height and body mass index in approximately 700000 individuals of European ancestry. *Hum. Mol. Genet.* **27**, 3641–3649 (2018).
55. Onat, F. & Cavdar, S. Cerebellar connections: hypothalamus. *Cerebellum* **2**, 263–269 (2003).
56. Zhu, J. N. & Wang, J. J. The cerebellum in feeding control: possible function and mechanism. *Cell Mol. Neurobiol.* **28**, 469–478 (2008).
57. Low, A. Y. T. et al. Reverse-translational identification of a cerebellar satiation network. *Nature* **600**, 269–273 (2021).
58. Floresco, S. B., Todd, C. L. & Grace, A. A. Glutamatergic afferents from the hippocampus to the nucleus accumbens regulate activity of ventral tegmental area dopamine neurons. *J. Neurosci.* **21**, 4915–4922 (2001).
59. Enard, W. et al. A humanized version of Foxp2 affects cortico-basal ganglia circuits in mice. *Cell* **137**, 961–971 (2009).
60. Demontis, D. et al. Discovery of the first genome-wide significant risk loci for attention deficit/hyperactivity disorder. *Nat. Genet.* **51**, 63–75 (2019).
61. Bulik-Sullivan, B. et al. An atlas of genetic correlations across human diseases and traits. *Nat. Genet.* **47**, 1236–1241 (2015).
62. Wei, H. S. et al. Erythrocytes are oxygen-sensing regulators of the cerebral microcirculation. *Neuron* **91**, 851–862 (2016).
63. Olusi, S. O. Obesity is an independent risk factor for plasma lipid peroxidation and depletion of erythrocyte cytoprotectic enzymes in humans. *Int. J. Obes. Relat. Metab. Disord.* **26**, 1159–1164 (2002).
64. Ozata, M. et al. Increased oxidative stress and hypozincemia in male obesity. *Clin. Biochem.* **35**, 627–631 (2002).
65. Druml, W., Laggner, A. N., Lenz, K., Grimm, G. & Schneeweiss, B. Pancreatitis in acute hemolysis. *Ann. Hematol.* **63**, 39–41 (1991).
66. Sakai, N. S., Taylor, S. A. & Chouhan, M. D. Obesity, metabolic disease and the pancreas—quantitative imaging of pancreatic fat. *Br. J. Radiol.* **91**, 20180267 (2018).
67. Peterson, C. G., Skoog, V. & Venge, P. Human eosinophil cationic proteins (ECP and EPX) and their suppressive effects on lymphocyte proliferation. *Immunobiology* **171**, 1–13 (1986).
68. Nakagome, K. et al. IL-5-induced hypereosinophilia suppresses the antigen-induced immune response via a TGF-beta-dependent mechanism. *J. Immunol.* **179**, 284–294 (2007).
69. Onyema, O. O. et al. Eosinophils downregulate lung alloimmunity by decreasing TCR signal transduction. *JCI Insight* **4**, e128241 (2019).
70. Yao, D. W., O'Connor, L. J., Price, A. L. & Gusev, A. Quantifying genetic effects on disease mediated by assayed gene expression levels. *Nat. Genet.* **52**, 626–633 (2020).

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Methods

Our research complies with all relevant ethnical regulations; no specific approval was needed.

TCSC regression

TCSC leverages the fact that the TWAS χ^2 statistic for a gene–tissue pair includes the direct effects of the gene on the disease, as well as the tagging effects of co-regulated tissues and genes with shared eQTLs or eQTLs in LD. Thus, genes that are co-regulated across many tissues will tend to have higher χ^2 statistics than genes regulated in a single tissue. TCSC determines that a tissue causally contributes to disease if genes with high co-regulation to the tissue have higher TWAS χ^2 statistics than genes with low co-regulation to the tissue.

We model the genetic component of gene expression as a linear combination of SNP-level effects:

$$W_{jgt'} = \sum_m X_{jm}\beta_{gt'm}, \tag{3}$$

where W_{jgt} is the *cis*-genetic component of gene expression in individual j for gene g and tissue t', X_{jm} is the standardized genotype of individual j for SNP m and $\beta_{gt'm}$ is the standardized effect of the mth SNP on the *cis*-genetic component of gene expression of gene g in tissue t'. We define the *cis*-genetic component of gene expression $W_{jgt'}$ to have mean 0 and variance 1 and $\beta_{gt'm}$ to have mean 0 and variance $\frac{1}{M_g}$, where M_g is the number of *cis* variants for gene g.

TCSC assumes that true gene–disease effects are i.i.d. across genes and tissues while accounting for the fact that *cis*-genetic components of gene expression (and *cis*-genetic predictions of gene expression) are correlated[1] (see Supplementary Figs. 12–15 for simulations where gene expression–trait effect sizes are not i.i.d. across genes and tissues; TCSC performs well despite violations of model assumptions). The high correlation of *cis*-eQTLs across tissues leads to tagging from co-regulated tissues[2]. We model phenotype as a linear combination of genetic components of gene expression across genes in different tissues:

$$Y_j = \sum_{t'}\sum_g W_{jgt}\alpha_{gt'} + \epsilon_j, \tag{4}$$

where Y_j is the (binary or continuous-valued) phenotype of individual j, α_{gt} is the standardized effect size of the *cis*-genetic component of gene expression on disease and ϵ_j is the component of phenotype not explained by *cis*-genetic components of gene expression. We emphasize that we model disease as a function of the unobserved true *cis*-genetic component of gene expression $W_{jgt'}$, not the genetically predicted value $\widehat{W}_{jgt'}$ obtained from gene expression prediction models. Equation (4) can be rewritten in terms of SNP-level effects:

$$Y_j = \sum_i X_{ji}\beta'_i + \sum_{t'}\sum_g\sum_m X_{jm}\beta_{gt'm}\alpha_{gt} + \epsilon_j, \tag{5}$$

where β'_i is a direct SNP–disease effect not mediated through gene expression.

We define the disease heritability explained by *cis*-genetic expression across all tissues as follows:

$$h^2_{\mathrm{ge}} = \mathrm{var}\left(\sum_{t'}\sum_g\sum_m X_{jm}\beta_{gt'm}\alpha_{gt}\right). \tag{6}$$

As W_{jgt} has mean 0 and variance 1 and α_{gt} is assumed to be i.i.d. across genes and tissues (see above), equation (5) implies that:

$$h^2_{\mathrm{ge}} = \sum_{t'}\sum_g \alpha^2_{gt'}, \tag{7}$$

analogous to the relationship between SNP effect sizes and SNP heritability[25]: $h^2_g = \mathrm{var}(\sum_i X_j\beta_i)$. We emphasize that the respective terms in equation (5) for each tissue t' are independent because $\alpha_{gt'}$ values are assumed to be i.i.d. across genes and tissues. It follows that the disease heritability explained by a particular tissue t' is:

$$h^2_{\mathrm{ge}(t')} = \mathrm{var}\left(\sum_g\sum_m X_{jm}\beta_{gt'm}\alpha_{gt'}\right), \tag{8}$$

which, given that $W_{jgt'}$ has mean 0 and variance 1 and $\alpha_{gt'}$ is i.i.d. across genes, reduces to:

$$h^2_{\mathrm{ge}(t')} = \sum_g \alpha^2_{gt'}. \tag{9}$$

Equations (7) and (9) imply that $h^2_{\mathrm{ge}} = \sum_{t'} h^2_{\mathrm{ge}(t')}$. Now, let $\alpha_{gt'}$ be a random variable drawn from a normal distribution with mean zero and tissue-specific variance $\mathrm{var}(\alpha_{gt'}) = \tau_{t'}$. Then:

$$h^2_{\mathrm{ge}(t')} = \sum_g \mathrm{var}(\alpha_{gt'}) = \tau_{t'}G_{t'}, \tag{10}$$

where $G_{t'}$ is the number of significantly *cis*-heritable genes in the model. In simulations, we demonstrate that, when there are similar numbers of *cis*-heritable genes across tissues, setting $G_{t'}$ to the total number of unique *cis*-heritable genes produces unbiased estimates in TCSC for the causal tissue; however, when there are varying numbers of *cis*-heritable genes across tissues (fewer in the causal tissue), this produces upward biased estimates (Supplementary Figs. 3 and 4) and thus setting $G_{t'}$ to the number of significantly *cis*-heritable genes in tissue t' is recommended. With this variance term, we can define a polygenic model that relates TWAS χ^2 statistics to co-regulation scores, which explicitly model the covariance structure of the χ^2 statistics. This strategy is analogous to modeling the dependence of GWAS χ^2 statistics on LD scores[25].

In a TWAS, the estimated value of the gene–disease effect size $\alpha_{gt'}$ is proportional to the correlation of the *cis*-genetic components of gene expression and their true gene–disease effect sizes for nearby genes across tissues, analogous to GCSC[22]:

$$E[\hat{\alpha}_{gt}] = \sum_{t'}\sum_{g'} r\left(\widehat{W}_{gt}, W_{g't'}\right)\alpha_{g't'} + \epsilon_g, \tag{11}$$

where $r(\widehat{W}_{gt}, W_{g't'})$ is the estimated correlation in *cis*-genetic predicted expression between gene g in tissue t and gene g' in tissue t'. ϵ_g is the component of phenotype not explained by *cis*-genetic components of gene expression, with mean 0 and variance σ^2_e/N.

The value of the TWAS χ^2 is proportional to the squared, estimated, disease-gene effect size and the GWAS sample size N as follows:

$$\chi^2_{gt} = N\hat{\alpha}^2_{gt} \tag{12}$$

Using equations (9) and (10), we can write the expectation of TWAS χ^2 as follows:

$$E\left[\chi^2_{gt}\right] = E\left[N\hat{\alpha}^2_{gt}\right] \tag{13}$$

$$= NE\left[\left(\sum_{t'}\sum_{g'}\hat{r}_{gg't'}\alpha_{g't'} + \epsilon_g\right)^2\right] \tag{14}$$

$$= N\sum_{t'}\sum_{g'}E\left[\hat{r}^2_{gg't'}\right]E\left[\alpha^2_{g't'}\right] + NE\left[\epsilon^2_g\right] \tag{15}$$

$$\approx N\sum_{t'}\sum_{g'}\left(r^2_{gg't'} + \frac{1}{N}\right)h^2_{\mathrm{ge}(t')}/G_{t'} + N\sigma^2_e/N \tag{16}$$

$$= N\sum_{t'}\sum_{g'}\left(r^2_{gg't'} + \frac{1}{N}\right)\tau_{t'} + \sigma^2_e \quad (17)$$

$$= N\sum_{t'}\sum_{g'} r^2_{gg't'}\tau_{t'} + \sum_{t'}\sum_{g'}\tau_{t'} + \sigma^2_e \quad (18)$$

$$= N\sum_{t'}\sum_{g'} r^2_{gg't'}\tau_{t'} + \sum_{t'}\mathrm{var}(\alpha_{g't'}) + \sigma^2_e \quad (19)$$

$$= N\sum_{t'} l(g,t;t')\,\tau_{t'} + 1 \quad (20)$$

$$= N\sum_{t'} l(g,t;t')\,h^2_{\mathrm{ge}(t')}/G_{t'} + 1. \quad (20.1)$$

To go from equation (15) to equation (16) we use the following relationship from the derivation of LDSC[13]:

$$E\left[\hat{r}^2_{gg't'}\right] \approx r^2_{gg't'} + \frac{1}{N}. \quad (21)$$

We go from equation (19) to equation (20) because the variance of the phenotype Y_j is $\sum_{t'}\mathrm{var}(\alpha_{g't'}) + \sigma^2_e$ and is equal to 1. We also introduce the notation that $\sum_{g'} r^2_{gg't'}$ is the tissue and gene co-regulation score $l(g,t;t')$, see below. We are interested in estimating $\tau_{t'}$, the per-gene disease heritability explained by the *cis*-genetic component of gene expression in tissue t'. From the derivation, the genome-wide tissue-specific contribution to disease heritability is estimated as:

$$h^2_{\mathrm{ge}(t')} = G_{t'}\tau_{t'}. \quad (22)$$

For the analysis of tissue-specific contributions to the covariance between two diseases, we can extend TCSC by using products of TWAS *z*-scores. Following the polygenic model described above, the expected product of TWAS *z*-scores in disease 1 and disease 2 for gene g and tagging tissue t is:

$$E\left[z^1_{gt}\times z^2_{gt}\right] = \sqrt{N_1N_2}\sum_{t'} l(g,t;t')\,\omega_{\mathrm{ge}(t')}/G_{t'} + \rho N_s/\sqrt{N_1N_2} \quad (23)$$

where N_1 is GWAS sample size for disease 1, N_2 is GWAS sample size for disease 2, t' indexes causal tissues, $l(g,t;t')$ is a tissue co-regulation score (see below), $\omega_{\mathrm{ge}(t')}$ is the genetic covariance explained by the *cis*-genetic component of gene expression in tissue t', $G_{t'}$ is the number of significantly *cis*-heritable genes in tissue t' (see below), ρ is the phenotypic correlation between disease 1 and disease 2 and N_s is the number of overlapping GWAS samples between disease 1 and disease 2. The last term represents the intercept[61] and, although we use a free intercept in the multivariate regression on co-regulation scores, the estimation of this term plays a role only in the estimation of regression weights (see below).

For estimates of $h^2_{\mathrm{ge}(t')}$ and $\omega_{\mathrm{ge}(t')}$, we use a free intercept; the estimation of $\rho N_s\sqrt{N_1N_2}$ serves only to inform the heteroscedasticity weights (see below) and is not used in the multivariate TCSC regression to estimate $\omega_{\mathrm{ge}(t')}$. To estimate s.e. values, we use a genomic block jackknife >200 genomic blocks with an equal number of genes in each. This is computed as the square root of the weighted variance across the jackknife estimates (where the weight of each block is equal to the sum of the regression weights for the genes in that block) multiplied by 200 blocks. We expect that the jackknife standard error will be conservative relative to the empirical standard error across estimates due to variation in causal signal across loci[71].

Estimating tissue co-regulation scores and correcting for bias

We define the co-regulation score of gene g with tissues t and t' as:

$$l(g,t;t') = \sum_{g'} r^2\left(\hat{W}_{g,t}, W_{g',t'}\right), \quad (24)$$

where W denotes the *cis*-genetic component of gene expression for a gene–tissue pair across individuals, $\hat{W}$ denotes the *cis*-predicted expression for a gene–tissue pair and genes g' are within ±1 Mb of the focal gene g. TCSC corrects for bias in tissue co-regulation scores arising from differences between *cis*-genetic versus *cis*-predicted expression (analogous to GCSC[22]). We apply bias correction to co-regulation scores in the special case when $g = g'$ and $t = t'$. Although co-regulation scores aim to estimate $r^2\left(\hat{W}_{g,t}, W_{g',t'}\right)$, the squared correlation of the predicted *cis*-genetic component of expression of gene g and tissue t (corresponding to the TWAS $\chi^2_{g,t}$ statistic) with the *actual cis*-genetic component of gene expression of gene g' in tissue t', when $g = g'$ and $t = t'$, the estimated value of $r^2\left(\hat{W}_{g,t}, W_{g',t'}\right)$ will always equal 1 because the estimate is based on $r^2\left(\hat{W}_{g,t}, \hat{W}_{g',t'}\right)$. However, this implies that predictions of the *cis*-genetic component of expression are perfectly accurate, which is unlikely to be the case. Therefore, the estimated value of $r^2\left(\hat{W}_{g,t}, W_{g',t'}\right)$ if left to equal 1 will cause co-regulation scores to be systematically inflated.

Therefore, when $g = g'$ and $t = t'$, we set:

$$r^2\left(\hat{W}_{g,t'}, W_{g',t'}\right) = R^2/h^2_{\mathrm{GCTA}}, \quad (25)$$

where R^2 is the cross-validation prediction statistic of the gene expression model for gene g in tissue t and h^2_{GCTA} is the GCTA-estimated *cis*-heritability of gene expression for gene g in tissue t. The quotient R^2/h^2_{GCTA} is the accuracy of the gene expression prediction model, which reflects the upper bound on how much the *cis*-predicted expression can be correlated with the true *cis*-genetic component of gene expression. Although we only consider genes with h^2_{GCTA} $P < 0.01$, the uncertainty in h^2_{GCTA} estimates should be modest and therefore not greatly impact our bias correction. We note that TWAS tests the null hypothesis that a specific weighted linear combination of SNPs is not associated with disease (and does not test the null hypothesis that the *cis*-genetic component of gene expression is not associated with disease).

TCSC regression weights

TCSC uses three sets of regression weights to increase power (analogous to GCSC[22]). The first regression weight is inversely proportional to $L(g,t)$, the total co-regulation score of each gene–tissue pair summed across tissues t':

$$L(g,t) = \sum_{t'}\sum\nolimits_{g'} r^2\left(\hat{W}_{g,t}, W_{g',t'}\right) \quad (26)$$

(without applying bias correction; see above), which allows TCSC to properly account for redundant contributions of co-regulated genes to TWAS χ^2 statistics.

The second regression weight is inversely proportional to $T(g,t)$, the number of tissues in which a gene is significantly *cis*-heritable:

$$T(g,t) = \sum_{t'\in g,t' \text{ significantly } cis\text{-heritable}} 1, \quad (27)$$

thereby upweighting signal from genes that are regulated in a limited number of tissues and preventing TCSC from attributing more weight to genes that are co-regulated across many tissues.

The third regression weight is inversely proportional to $H_{h^2}(g,t)$, the heteroscedasticity of χ^2 statistics, and is computed differently for estimates of $h^2_{\mathrm{ge}(t')}$ than for estimates of $\omega_{\mathrm{ge}(t')}$ (analogous to GCSC[22] and cross-trait LDSC[61], respectively).

For estimates of $h^2_{\mathrm{ge}(t')}$, we estimate $H_{h^2}(g,t)$ in two steps. First, we make a crude estimate of heritability explained by predicted expression (h^2_{ge}) as follows:

$$\mu_\chi = N\mu_L h^2_{\text{ge}} + 1, \tag{28}$$

where μ_χ is the mean χ^2 statistic:

$$\mu_\chi = \frac{\sum_{t'} \sum_{g'} \chi^2_{g',t'}}{\sum_{t'} G_{t'}}, \tag{29}$$

where N is the GWAS sample size, g' iterates over significantly *cis*-heritable genes, t' iterates over tissues and μ_L is the mean value of total co-regulation across tissues t':

$$\mu_L = \frac{\sum_{t'} \sum_{g'} L(g', t')}{\sum_{t'} G_{t'}}. \tag{30}$$

Then, we compute the heteroscedasticity for each significantly *cis*-heritable gene–tissue pair as:

$$H_{h^2}(g,t) = \left(NL(g,t)\, h^2_{\text{ge}} + 1\right)^2. \tag{31}$$

Finally, we combine the three regression weights as follows:

$$\text{Weight}_{h^2}(g,t) = \frac{1}{L(g,t)\, T(g,t)\, H_{h^2}(g,t)}. \tag{32}$$

For estimates of $\omega_{\text{ge}(t')}$, we estimate $H_\omega(g,t)$ in two steps. First, we regress the products of TWAS z-scores on total tissue co-regulation scores, $L(g,t)$, using regression weights, $\text{Weight}_{\omega(g,t)}$, computed as follows:

$$\text{Weight}_\omega(g,t) = \frac{1}{L(g,t)\, T(g,t)\, H_\omega(g,t)} \tag{33}$$

where $H_\omega(g,t)$ is first estimated as follows:

$$H_\omega(g,t) = \left(N_1 L(g,t)\, h^2_{\text{ge}}(\text{trait 1}) + 1\right)\left(N_2 L(g,t)\, h^2_{\text{ge}}(\text{trait 2}) + 1\right) + \left(\frac{\sqrt{N_1 N_2}\,\omega_{\text{ge}} L(g,t)}{\sum_{t'} \frac{G_{t'}}{T'}}\right)^2, \tag{34}$$

where $h^2_{ge}(\text{trait 1})$ is the crude heritability estimate for trait 1 and $h^2_{\text{ge}}(\text{trait 2})$ is the crude heritability estimate for trait 2, ω_{ge} is estimated as $\frac{\sum_{t'} \sum_{g't'} z^{(1)}_{g't'} z^{(2)}_{g't'}}{\sqrt{N_1 N_2}}$, N_1 is the sample size of the first GWAS, N_2 is the sample size of the second GWAS and T' is the total number of tissues in the regression.

Second, we use the regression intercept to estimate the product ρN_s:

$$\rho N_s = \text{intercept} \times \sqrt{N_1 N_2}, \tag{35}$$

where ρ represents the phenotypic correlation between trait 1 and trait 2 and N_s represents the number of shared samples between GWAS 1 and GWAS 2. We also use the coefficient of the regression to update our estimate of ω_{ge}, such that we may update the heteroscedasticity weight as follows:

$$H_\omega(g,t) = \left(N_1 L(g,t)\, h^2_{\text{ge}}(\text{trait 1}) + 1\right)\left(N_2 L(g,t)\, h^2_{\text{ge}}(\text{trait 2}) + 1\right) + \left(\frac{\sqrt{N_1 N_2}\,\omega_{\text{ge}} L(g,t)}{\sum_{t'} \frac{G_{t'}}{T'}} + \rho N_s \sqrt{N_1 N_2}\right)^2. \tag{36}$$

Finally, we combine the three regression weights as follows:

$$\text{Weight}_\omega(g,t) = \frac{1}{L(g,t)\, T(g,t)\, H_\omega(g,t)}. \tag{37}$$

Gene expression prediction models and tissue co-regulation scores in GTEx data

To construct gene expression prediction models, we applied FUSION[21] (Code availability) to individual-level GTEx data by regressing measured gene expression on genotypes of common variants (MAF > 0.05) and covariates provided by GTEx[19]. FUSION uses several different regression models: single eQTL, elastic net, lasso and BLUP, and the following covariates: sex, five genotyping principal components, PEER (probabilistic estimation of expression residuals) factors[72] and assay type. We defined significantly *cis*-heritable genes as protein-coding genes with GCTA heritability $P < 0.01$ (ref. 21), heritability estimate >0 and adjusted $R^2 > 0$ in cross-validation prediction.

We used gene expression prediction models of significantly *cis*-heritable genes to predict expression into 489 European individuals from 1000 Genomes[27]. We then estimated tissue co-regulation scores using equations (24) and (25)[73], where *cis*-predicted gene expression is used to estimate the *cis*-genetic component of gene expression.

GWAS summary statistics and TWAS association statistics

We collected GWAS summary statistics from 78 relatively independent heritable complex diseases and traits (average n = 302,000) with heritability z-score >6, including 33 diseases/traits from UK Biobank[74]. We estimated the heritability of all summary statistics and genetic correlation of all pairs of summary statistics. We excluded traits with heritability z-score <6, using S-LDSC with the baseline-LD v.2.2 model[13,28,29] and as done previously[28]. We excluded one of each pair of traits that are both genetically correlated and have significantly overlapping samples. Specifically, for any pair of non-UK Biobank traits with an estimated sample overlap greater than the following threshold-squared, cross-trait, LDSC intercept/(trait 1 S-LDSC intercept × trait 2 S-LDSC intercept) > 0.1[61], the trait with the larger SNP heritability z-score was retained. For any pair of UK Biobank traits with a squared genetic correlation >0.1, the trait with the larger SNP heritability z-score was retained[75]. In total, this procedure resulted in 78 sets of relatively independent GWAS summary statistics. We limited all analyses (including cross-trait analyses) to the 78 relatively independent traits to avoid redundant findings across single-trait (and cross-trait) analyses. For the brain-specific analysis, we first selected brain-related diseases and complex traits, for example, psychiatric disorders and behavioral phenotypes, excluding multi-case–control studies and case versus case studies. Then, we applied our standard filters as described above, but relaxing the threshold of squared genetic correlation to 0.25.

We used FUSION[21] (Code availability) to compute TWAS association statistics for each pair of signed GWAS summary statistics and each significantly *cis*-heritable gene–tissue pair, across the two scenarios described above. We restrict gene expression prediction models and TWAS association statistics for each tissue to significantly *cis*-heritable genes in that tissue, defined as genes with significantly positive *cis*-heritability (two-sided $P < 0.01$; estimated using GCTA[76]) and positive adjusted R^2 in cross-validation prediction. We further removed genes within the major histocompatibility complex (chromosome 6, 29–33 Mb) and TWAS $\chi^2 > 80$ or $\chi^2 > 0.001N$, where N is the GWAS sample size, as previously used for quality control in the heritability analysis of GWAS summary statistics[13]. TCSC scales linearly with the number of genes and quadratically with the number of tissues. After all input datasets are created and processed, running TCSC on a single real GWAS trait with 39 tissues takes about 2 min[73]. We employ a per-trait FDR (as in refs. 77,78) rather than a global FDR (as in ref. 7), because power is likely to vary across traits and there are a sufficiently large number of independent quantities estimated per trait ($\pi_{t'}$ jointly estimated across 39 tissues); a global FDR is more appropriate when there are far fewer independent quantities estimated per trait, for example, due to nonindependent, marginal tissue associations in ref. 7 For cross-trait analysis, we estimated the genetic correlation of each of 3,003 pairs of the 78 disease/traits analyzed in the present study; the

262 pairs analyzed by cross-trait analysis had significantly nonzero genetic correlation ($P < 0.05/3{,}003$).

RTC Coloc and LDSC-SEG analysis of GWAS summary statistics and GTEx tissues

We downloaded supplementary tables for the RTC Coloc method[2] and for LDSC-SEG[7]. For traits in our set of 78 GWAS summary statistics that were not analyzed by the LDSC-SEG study and for traits that are inherently brain related (because these traits require a different procedure for generating tissue-specific gene sets), we ran LDSC-SEG ourselves. To this end, we downloaded LD scores for GTEx tissues and specifically expressed gene set SNP-level annotations (https://alkesgroup.broadinstitute.org/LDSCORE/LDSC_SEG_ldscores) and ran LDSC-SEG[7] using S-LDSC v.1.0.0. For brain-related traits, we additionally ran a brain-specific analysis using LDSC-SEG[7]. Briefly, specifically expressed genes were determined via a Student's *t*-test of the sentinel brain tissue against all other brain tissues, rather than against all other nonbrain GTEx tissues, as done in the primary analysis of the LDSC-SEG study. For traits in our set that were not analyzed by the RTC Coloc study, of which there were few, we did not apply their method, because it was too computationally intensive to apply to real trait data.

Statistics and reproducibility

First, as described above, we downsampled or meta-analyzed GTEx tissues to reduce the potential for type I error in our analysis; downsampling was performed randomly. Second, as described above, GWAS summary statistics with an SNP heritability *z*-score <6 were excluded, due to the potential to have low power. Third, in analyses of real diseases/traits, we verified that we obtained consistent results when analyzing different GWAS cohorts. We obtained independent GWAS summary statistics for 10 traits implicated in 13 significant tissue–trait pairs (Supplementary Table 4) and confirmed the same direction of effect for 13 of 13 tissue–trait pairs (including FDR < 5% for 7 of 13 tissue–trait pairs and FDR < 10% for 9 of 13 tissue–trait pairs) (Supplementary Table 5). Fourth, 1,000 independent genetic architectures were simulated to robustly evaluate TCSC performance. Randomization and blinding were not pertinent to our study.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

We have made 78 GWAS summary statistics and 41 brain-specific summary statistics publicly available at https://github.com/TiffanyAmariuta/TCSC/tree/main/sumstats (https://doi.org/10.5281/zenodo.8030594)[73], gene expression prediction models publicly available at https://alkesgroup.broadinstitute.org/TCSC/GeneExpressionModels, TWAS association statistics publicly available at https://alkesgroup.broadinstitute.org/TCSC/TWAS_sumstats, tissue co-regulation scores publicly available at https://github.com/TiffanyAmariuta/TCSC/tree/main/coregulation_scores and TCSC output publicly available at https://github.com/TiffanyAmariuta/TCSC/tree/main/results. Gene expression and genotype data were acquired from the GTEx v.8 eQTL dataset (dbGaP, accession no. phs000424.v8.p2) and 1000 Genomes phase 3 data were downloaded from https://data.broadinstitute.org/alkesgroup/FUSION/LDREF.tar.bz2. Source data are provided with this paper.

Code availability

TCSC software including a quick start tutorial is available at https://github.com/TiffanyAmariuta/TCSC/ (https://doi.org/10.5281/zenodo.8030594)[73]. The Mancuso Lab TWAS Simulator is available at https://github.com/mancusolab/twas_sim. The FUSION software is available at http://gusevlab.org/projects/fusion. The simulation code for RTC Coloc is available at https://github.com/TiffanyAmariuta/TCSC/tree/main/simulation_analysis.

References

71. Weissbrod, O. et al. Leveraging fine-mapping and multipopulation training data to improve cross-population polygenic risk scores. *Nat. Genet.* **54**, 450–458 (2022).
72. Stegle, O., Parts, L., Durbin, R. & Winn, J. A Bayesian framework to account for complex non-genetic factors in gene expression levels greatly increases power in eQTL studies. *PLoS Comput. Biol.* **6**, e1000770 (2010).
73. Amariuta, T. TCSC GitHub Repository (2023); https://doi.org/10.5281/zenodo.8030594
74. Bycroft, C. et al. The UK Biobank resource with deep phenotyping and genomic data. *Nature* **562**, 203–209 (2018).
75. Gazal, S. et al. Functional architecture of low-frequency variants highlights strength of negative selection across coding and non-coding annotations. *Nat. Genet.* **50**, 1600–1607 (2018).
76. Yang, J., Lee, S. H., Goddard, M. E. & Visscher, P. M. GCTA: a tool for genome-wide complex trait analysis. *Am. J. Hum. Genet.* **88**, 76–82 (2011).
77. Reshef, Y. A. et al. Detecting genome-wide directional effects of transcription factor binding on polygenic disease risk. *Nat. Genet.* **50**, 1483–1493 (2018).
78. Zhang, M. J. et al. Polygenic enrichment distinguishes disease associations of individual cells in single-cell RNA-seq data. *Nat. Genet.* **54**, 1572–1580 (2022).

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Author contributions

T.A. and A.L.P. conceived and designed the study. T.A. conducted simulation and real data analysis. K.S.-W. helped design the simulation framework. T.A. and A.L.P. wrote the initial draft of the manuscript. All authors contributed to the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41588-023-01474-z.

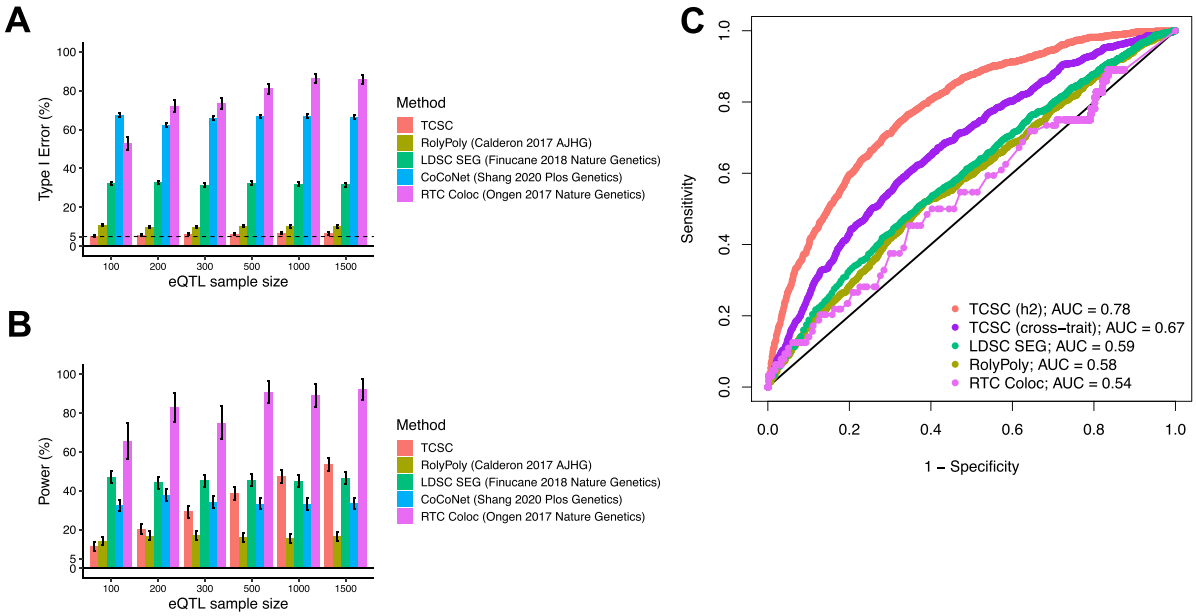
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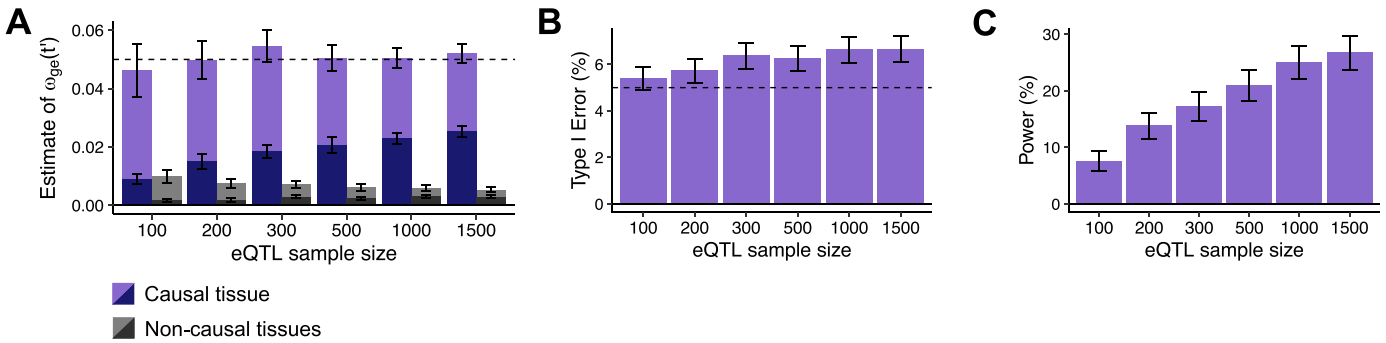
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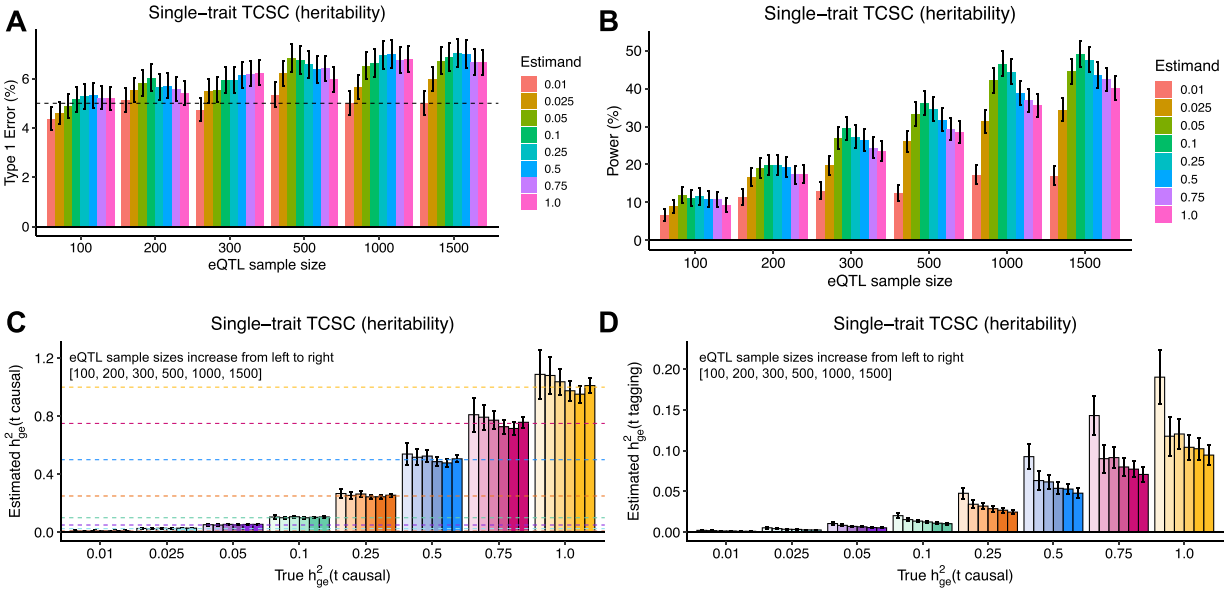
Extended Data Fig. 1 | Comparison of tissue-trait association methods with TCSC in simulations. (**a**) Percentage of estimates of $h^2_{ge(t')}$ for noncausal tissues that were significantly positive at $P < 0.05$. $h^2_{ge(t\,causal)}$ is set to 10% and GWAS sample size is set to 10,000. (**b**) Percentage of estimates of $h^2_{ge(t')}$ for causal tissues that were significantly positive at $p < 0.05$. In panels A and B, we performed n = 1,000 independent simulated genetic architectures for TCSC, LDSC-SEG, CoCoNet, and RolyPoly and n = 100 independent simulated genetic architectures for RTC Coloc due to the computationally intensive nature of the method across each eQTL sample size (n = 100, 200, 300, 500, 1000, 1500); we used a one-sided z-test and the genomic block jackknife standard error to obtain p-values and data are presented as mean values ± 1.96 × s.e.m. (**c**) Receiver operating characteristic (ROC) curves for each method, including cross-trait TCSC, across 1,000 uniformly spaced p-values between 0 and 1 used as the threshold to identify a causal tissue at a simulation eQTL sample size of 300, most closely matching real data analysis. We note that CoCoNet cannot be compared here because the method identifies the single most likely causal tissue using maximum likelihood estimation rather than via p-value. Numerical results are provided in Supplementary Tables 1, 2.

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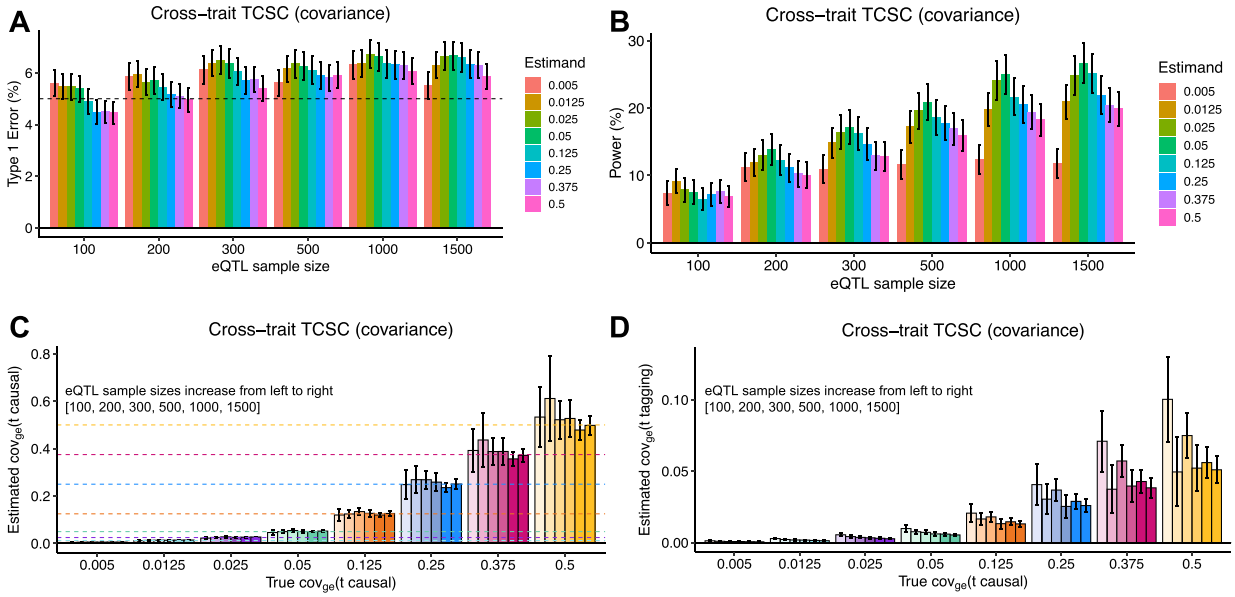
Extended Data Fig. 2 | Robustness and power of cross-trait TCSC in simulations. (**a**) Bias in estimates of genetic covariance explained by the *cis*-genetic component of gene expression in tissue t' ($\omega_{ge(t't')}$) for causal (purples) and noncausal (grays) tissues, across n = 1,000 independent simulations per eQTL sample size (n = 100, 200, 300, 500, 1000, 1500). Light purple (resp. gray) indicates that $G_{t't}$ was set to the total number of true *cis*-heritable genes across tissues, dark purple (resp. gray) indicates that $G_{t't}$ was set to the number of significantly *cis*-heritable genes detected in each tissue. The dashed line indicates the true value of $\omega_{ge(t't)}$ for causal tissues. (**b**) Percentage of estimates of $\omega_{ge(t't)}$ for noncausal tissues that were significantly positive at $p < 0.05$. (**c**) Percentage of estimates of $\omega_{ge(t't)}$ for causal tissues that were significantly positive at $p < 0.05$. For panels B and C, we performed n = 1,000 independent simulations per eQTL sample size (n = 100, 200, 300, 500, 1000, 1500) and used a one-sided z-test to obtain p-values. In all panels, data are presented as mean values ± 1.96 × s.e.m. Numerical results are reported in Supplementary Table 3.

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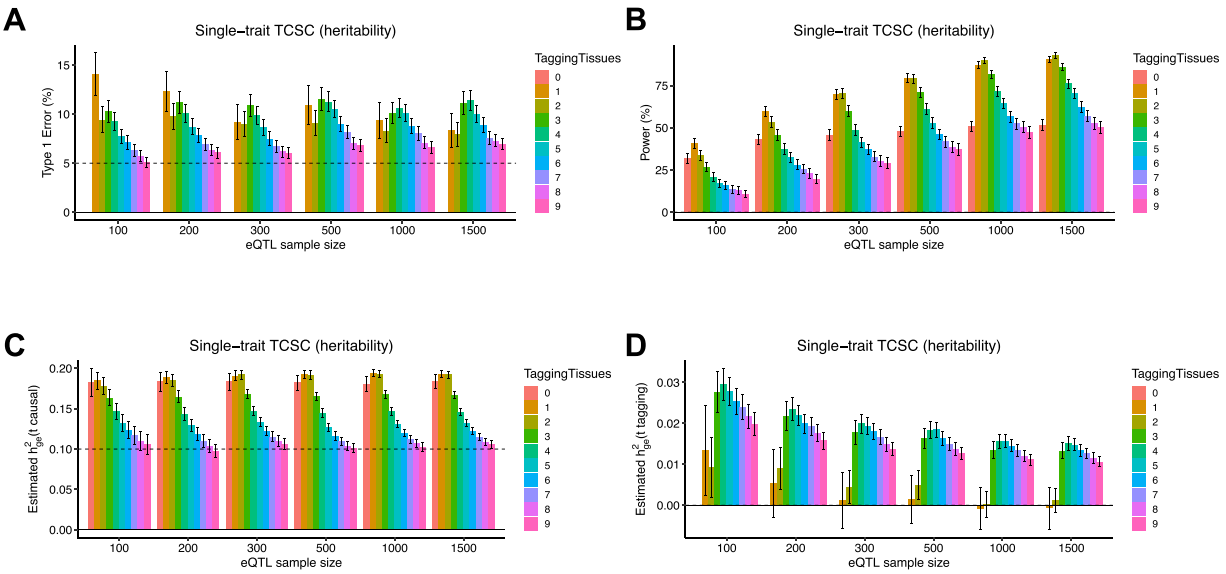
Extended Data Fig. 3 | Robustness and power of TCSC regression in simulations with different values of $h^2_{ge(t\prime)}$. (**a**) Type I error per true value of $h^2_{ge(t\prime)}$ in the causal tissue. False positive event is defined as $h^2_{ge(t\prime)} > 0$ for noncausal tissues at $p < 0.05$. (**b**) Power to detect the causal tissue per true value of $h^2_{ge(t\prime)}$ in the causal tissue. A true positive event is defined as $h^2_{ge(t\prime)} > 0$ for causal tissues at $p < 0.05$. (**c**) Bias on causal estimates of $h^2_{ge(t\prime)}$ for different true values of the causal tissue $h^2_{ge(t\prime)}$. Dashed lines indicate true values of $h^2_{ge(t\prime)}$. (**d**) Bias on noncausal estimates of $h^2_{ge(t\prime)}$ for different true values of the causal tissue $h^2_{ge(t\prime)}$. In all panels, we performed n = 1,000 independent simulated genetic architectures across different eQTL sample sizes (n = 100, 200, 300, 500, 1000, 1500); we used a one-sided z-test and the genomic block jackknife standard error to obtain p-values and data are presented as mean values ± 1.96 × s.e.m. The value of $G_{t\prime}$ is set to the total number of unique *cis*-heritable genes across all tissues.

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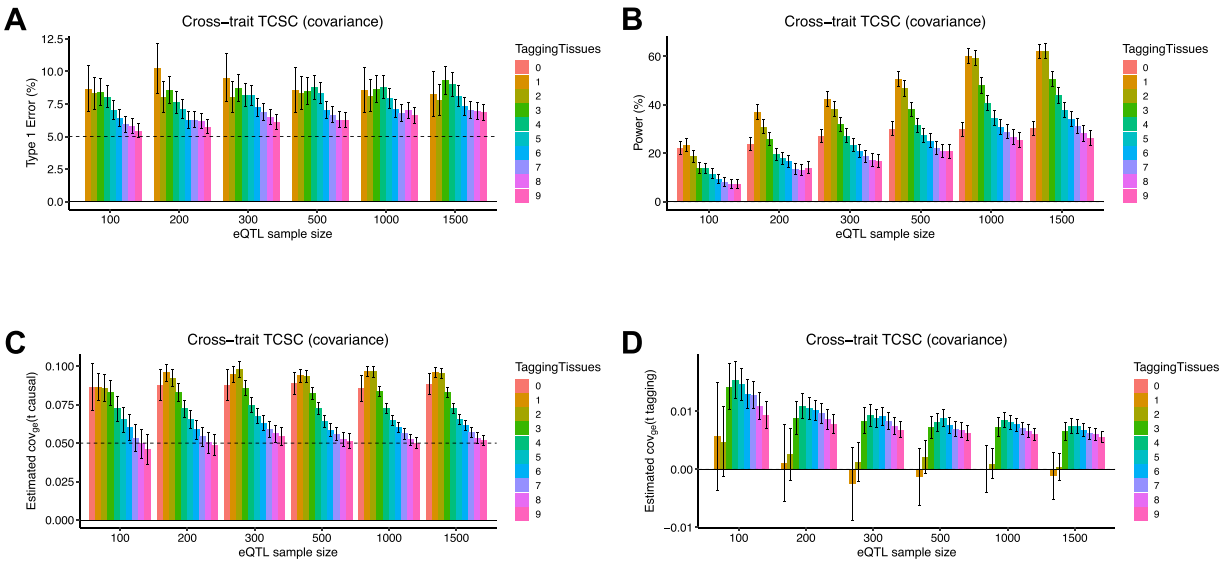
Extended Data Fig. 4 | Robustness and power of cross-trait TCSC regression in simulations with different values of $\omega_{ge(t\prime)}$. (**a**) Type I error per true value of $\omega_{ge(t\prime)}$ in the causal tissue. False positive event is defined as $\omega_{ge(t\prime)} > 0$ for noncausal tissues at $p < 0.05$. (**b**) Power to detect the causal tissue per true value of $\omega_{ge(t\prime)}$ in the causal tissue. A true positive event is defined as $\omega_{ge(t\prime)} > 0$ for causal tissues at $p < 0.05$. (**c**) Bias on causal estimates of $\omega_{ge(t\prime)}$ for different true values of the causal tissue $\omega_{ge(t\prime)}$. Dashed lines indicate true values of $\omega_{ge(t\prime)}$. (**d**) Bias on noncausal estimates of $\omega_{ge(t\prime)}$ for different true values of the causal tissue $\omega_{ge(t\prime)}$. In all panels, we performed n = 1,000 independent simulated genetic architectures across different eQTL sample sizes (n = 100, 200, 300, 500, 1000, 1500); we used a one-sided z-test and the genomic block jackknife standard error to obtain p-values and data are presented as mean values ± 1.96 × s.e.m. The value of $G_{t\prime}$ is set to the total number of unique *cis*-heritable genes across all tissues.

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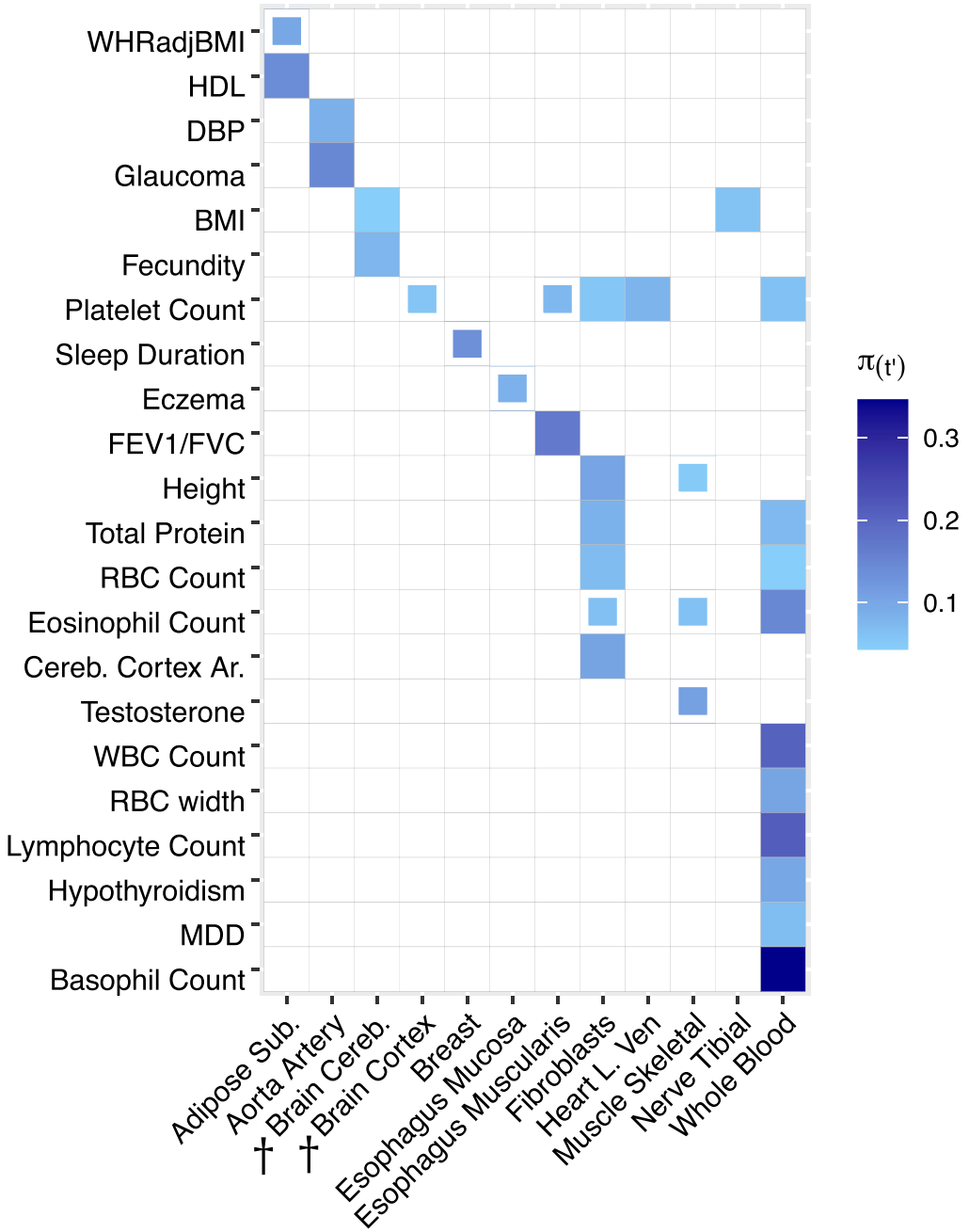
Extended Data Fig. 5 | Robustness and power of TCSC regression in simulations with different numbers of noncausal tissues. (**a**) Type I error involving a variable number of noncausal tissues in the presence of a single causal tissue. False positive event defined as $h^2_{ge(t\prime)} > 0$ for noncausal tissues at $p < 0.05$. Note, when there are 0 tagging tissues, there is no measurement of type I error. (**b**) Power to detect the causal tissue in which $h^2_{ge(t\prime)} > 0$ for causal tissues at $p < 0.05$. (**c**) Bias on estimates of $h^2_{ge(t\prime)}$ for the causal tissue, while changing the number of noncausal tissues in the model. The dashed line indicates that the true value of $h^2_{ge(t\prime)}$. (**d**) Bias on estimates of $h^2_{ge(t\prime)}$ for noncausal tissues, while changing the number of noncausal tissues in the model. In all panels, we performed n = 1,000 independent simulated genetic architectures across different eQTL sample sizes (n = 100, 200, 300, 500, 1000, 1500); we used a one-sided z-test and the genomic block jackknife standard error to obtain p-values and data are presented as mean values ± 1.96 × s.e.m. The value of $G_{t\prime}$ is set to the total number of unique *cis*-heritable genes across all tissues.

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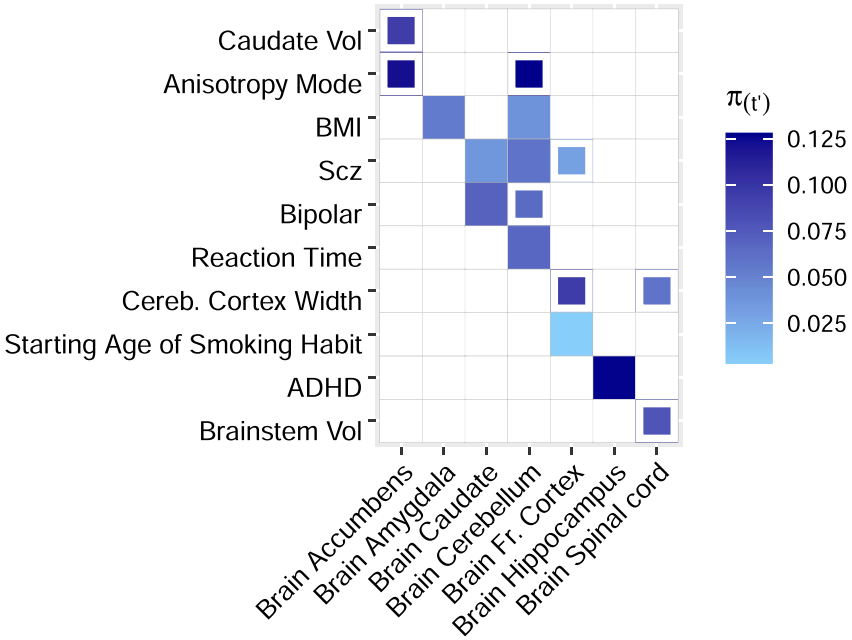
Extended Data Fig. 6 | Robustness and power of cross-trait TCSC regression in simulations with different numbers of noncausal tissues. (**a**) Type I error involving a variable number of noncausal tissues in the presence of a single causal tissue. False positive event defined as $\omega_{ge(t\prime)} > 0$ for noncausal tissues at $p < 0.05$. Note, when there are 0 tagging tissues, there is no measurement of type I error. (**b**) Power to detect the causal tissue in which $\omega_{ge(t\prime)} > 0$ for causal tissues at $p < 0.05$. (**c**) Bias on estimates of $\omega_{ge(t\prime)}$ for the causal tissue, while changing the number of noncausal tissues in the model. The dashed line indicates that the true value of $\omega_{ge(t\prime)}$. (**d**) Bias on estimates of $\omega_{ge(t\prime)}$ for noncausal tissues, while changing the number of noncausal tissues in the model. In all panels, we performed n = 1,000 independent simulated genetic architectures across different eQTL sample sizes (n = 100, 200, 300, 500, 1000, 1500); we used a one-sided *z*-test and the genomic block jackknife standard error to obtain p-values and data are presented as mean values ± 1.96 × s.e.m. The value of $G_{t\prime}$ is set to the total number of unique *cis*-heritable genes across all tissues.

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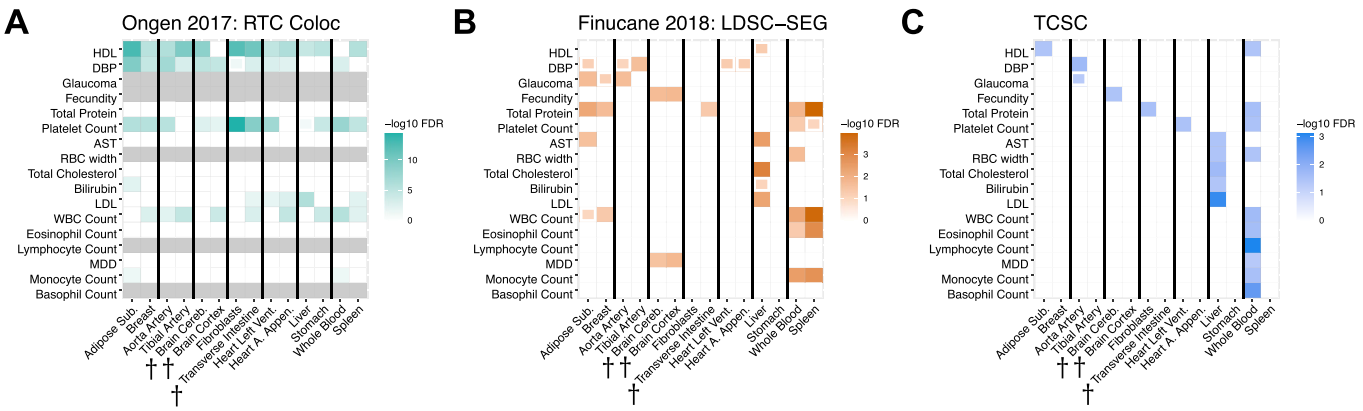
Extended Data Fig. 7 | Tissue-specific contributions to disease and complex trait heritability in secondary analysis of 23 tissues, removing tissues with small eQTL sample size. We report estimates of the proportion of disease heritability explained by the *cis*-genetic component of gene expression in tissue t' ($\pi_{t'}$). Results are shown for tissue-trait pairs with FDR <= 10%; full boxes denote FDR of 5% or lower and partial boxes denote FDR between 5% and 10%. Tissues are ordered alphabetically. Color corresponds to $\pi_{t'}$, the proportion of common variant heritability causally explained by predicted gene expression in tissue t'. These results are largely consistent with the analysis of 39 GTEx tissues (Fig. 4). WHRadjBMI: waist-hip-ratio adjusted for body mass index. HDL: high-density lipoprotein. DBP: diastolic blood pressure. BMI: body mass index. FEV_1/FVC: forced expiratory volume in one second divided by forced vital capacity. Cereb. Cortex Ar.: cerebral cortex surface area. WBC Count: white blood cell count. RBC Count: red blood cell count. MDD: major depressive disorder. Daggers next to a tissue indicate a meta-tissue.

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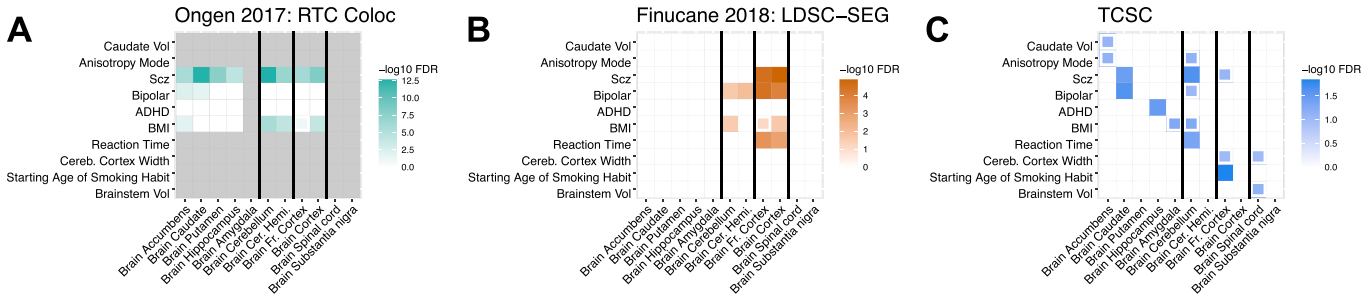
Extended Data Fig. 8 | Tissue-specific contributions to disease and complex trait heritability in brain-specific analysis. We separately analyzed results for 41 brain-related diseases and complex traits and 13 brain tissues. Results are shown for tissue-trait pairs with FDR <= 10%; full boxes denote FDR of 5% or lower and partial boxes denote FDR between 5% and 10%. Tissues are ordered alphabetically. Each tissue has an eQTL sample size ranging from 101 to 189 individuals. Color corresponds to $\pi_{t'}$, the proportion of common variant heritability causally explained by predicted gene expression in tissue t'. Caudate Vol: caudate volume. BMI: body mass index. Scz: schizophrenia. Bipolar: bipolar disorder. Brainstem Vol: brainstem volume. Cereb. Cortex Width: cerebral cortex width. ADHD: attention-deficit/hyperactivity disorder. Brainstem Vol: brainstem volume .

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Extended Data Fig. 9 | Comparison of disease-critical tissues identified by RTC Coloc, LDSC-SEG and TCSC for all 17 disease/traits with causal tissue-trait associations identified by TCSC. FDR significance of trait-tissue associations across three different methods for each of 17 traits with a significant tissue found by TCSC from Fig. 3 and 7 causal tissues, plus each tissue's most highly genetically correlated GTEx tissue. Full boxes denote FDR of 5% or lower and partial boxes denote FDR between 5% and 10%. Thicker black lines separate the causal tissue found in the primary TCSC analysis (left) from its most highly genetically correlated GTEx tissue (right), with two exceptions. First, breast tissue was the most highly genetically correlated tissue for two causal tissues, adipose subcutaneous and thyroid; therefore, these three tissues appear as a trio. Second, the aorta artery and tibial artery are each other's most highly genetically correlated tissue and both are a causal tissue different traits by TCSC. (**a**) RTC Coloc (Ongen 2017 Nat Genet), (**b**) LDSC-SEG (Finucane 2018 Nat Genet), (**c**) TCSC. Per-trait FDR in panels A and C, FDR across traits and tissues in panel B. WHRadjBMI: waist-hip-ratio conditional on body mass index. HDL: high-density lipoprotein. DBP: diastolic blood pressure. BMI: body mass index. FEV1/FVC: forced expiratory volume in one second divided by forced vital capacity. Cereb. Cortex Ar.: cerebral cortex surface area. AST: aspartate aminotransferase. LDL: low-density lipoprotein. WBC: white blood cell count. MDD: major depressive disorder. Daggers next to a tissue indicate a meta-tissue. For BMI, fecundity, and cereb. cortex ar., LDSC-SEG brain-specific analysis results are used.

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Extended Data Fig. 10 | Comparison of disease-critical tissues identified by RTC Coloc, LDSC-SEG and TCSC for brain-specific analysis. a–**c**, Brain-specific tissue-trait pairs identified by three methods: RTC Coloc (**a**), LDSC-SEG (**b**) and TCSC (**c**). Full boxes denote FDR of 5% or lower and partial boxes denote FDR between 5% and 10%. Each tissue has an eQTL sample size ranging from 100 to 189 individuals. Additionally, we include each tissue's most highly genetically correlated GTEx tissue. Thicker black lines separate the causal tissue found in the primary TCSC analysis (left) from its most highly genetically correlated GTEx tissue (right), with one exception: brain accumbens, caudate, putamen, hippocampus, and amygdala are all highly genetically correlated, with no pairs of exclusively high genetic correlation. Caudate Vol: caudate volume. BMI: body mass index. Scz: schizophrenia. Bipolar: bipolar disorder. Brainstem Vol: brainstem volume. Cereb. Cortex Width: cerebral cortex width. ADHD: attention-deficit/hyperactivity disorder. Brainstem Vol: brainstem volume.

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☐	☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
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☐	☒ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
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☐	☒ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No software was used for data collection, as there was no data collection in this study.
Data analysis	Open source: S-LDSC v1.0.0 (Finucane 2015 Nat Genet) for estimation of trait heritability, cross-trait LDSC v1.0.0 (Bulik-Sullivan, Finucane 2015 Nat Genet) for estimation of genetic correlation and covariance for pairs of traits, FUSION (most recently updated May 24, 2022; no version number available) (Gusev 2016 Nat Genet) for construction of gene expression prediction models and transcriptome-wide association studies, TCSC (custom code; https://github.com/TiffanyAmariuta/TCSC/tree/main/analysis) for the implementation of the method describe in this paper, LDSC SEG using S-LDSC v1.0.0 (Finucane 2018 Nat Genet) for analysis of specifically expressed genes, RTC Coloc (Ongen 2017 Nat Genet) using custom implementation to accommodate simulation framework (custom code; https://github.com/TiffanyAmariuta/TCSC/tree/main/simulation_analysis) for identifying trait-relevant tissues using colocalization analysis, RolyPoly v0.1.0 (Calederon 2017 AJHG) for identifying trait-relevant tissues in simulation analysis, CoCoNet v1.0 (Shang 2020 Plos Genet) for identifying trait-relevant tissues in simulation analysis, TWAS simulation framework (https://github.com/mancusolab/twas_sim; no version number available), R v3.6.1, python 3.6.2

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

No raw data was collected for this manuscript. GWAS summary statistics were previously analyzed by referenced studies and were deposited here (https://github.com/TiffanyAmariuta/TCSC/tree/main/sumstats). We have made publicly available the following intermediate files at the following paths: TWAS summary statistics (https://alkesgroup.broadinstitute.org/TCSC/TWAS_sumstats/), gene expression prediction models (https://alkesgroup.broadinstitute.org/TCSC/GeneExpressionModels/), tissue co-regulation scores (https://github.com/TiffanyAmariuta/TCSC/tree/main/coregulation_scores). Gene expression and genotype data was acquired from the GTEx v8 eQTL dataset (dbGaP Accession phs000424.v8.p2) and 1000 Genomes phase 3 data (https://data.broadinstitute.org/alkesgroup/FUSION/LDREF.tar.bz2).

Human research participants

Policy information about studies involving human research participants and Sex and Gender in Research.

Reporting on sex and gender	NA
Population characteristics	NA
Recruitment	NA
Ethics oversight	NA

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	eQTL sample sizes across GTEx tissues were variable. While the calibration of our method relies on similar sample sizes across jointly modeled tissues, our primary analysis reports results after downsampling each GTEx tissue to the average GTEx tissue sample size (N = 320 individuals) or meta-analyzing smaller and genetically correlated tissues to obtain a meta-sample size of 320 individuals. This approach maximizing the number of tissues that could be fairly analyzed in our study.
Data exclusions	We only discarded data from the kidney cortex which had fewer than 100 individuals, as results were expected to be unreliable. GWAS summary statistics with a SNP heritability z-score < 6 were excluded, due to the potential to have low power.
Replication	In analyses of real diseases/traits, we verified that we obtained consistent results when analyzing different GWAS cohorts. We obtained independent GWAS summary statistics for 10 traits implicated in 13 significant tissue-trait pairs (Supplementary Table 4) and confirmed the same direction of effect for 13 of 13 tissue-trait pairs (including FDR < 5% for 7 of 13 tissue-trait pairs, FDR < 10% for 9 of 13 tissue-trait pairs) (Supplementary Table 5). Each simulation of TCSC was performed independently (e.g. independent genetic architectures); there were 1,000 independent results for every simulation result reported.
Randomization	For real data analysis, we only analyzed GTEx data. Covariates for GTEX individuals (age, sex, genotyping PCs, gene expression PEER factors, sequencing tech) were used in the construction of gene expression prediction models as previously analyzed by the GTEx consortium to identify eQTLs and by Gusev 2016 Nat Genet via the algorithm of TWAS. For the subsampling of individuals per GTEx tissue, this process was done randomly, e.g. without any knowledge of individual-level covariates. For simulation analysis, individual genotypes were simulated from real people and LD and thus there was no randomization to be performed and there were no individual-level covariates of interest.
Blinding	The investigators were not blinded to any analysis or outcome assessment, as this is not a case/control study.

Reporting for specific materials, systems and methods

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Materials & experimental systems

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☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging